

Source Separation 1

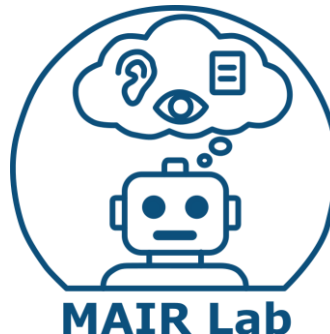
안인규 (Inkyu An)

Speech And Audio Recognition
(오디오 음성인식)

<https://mairlab.kookmin.ac.kr/>

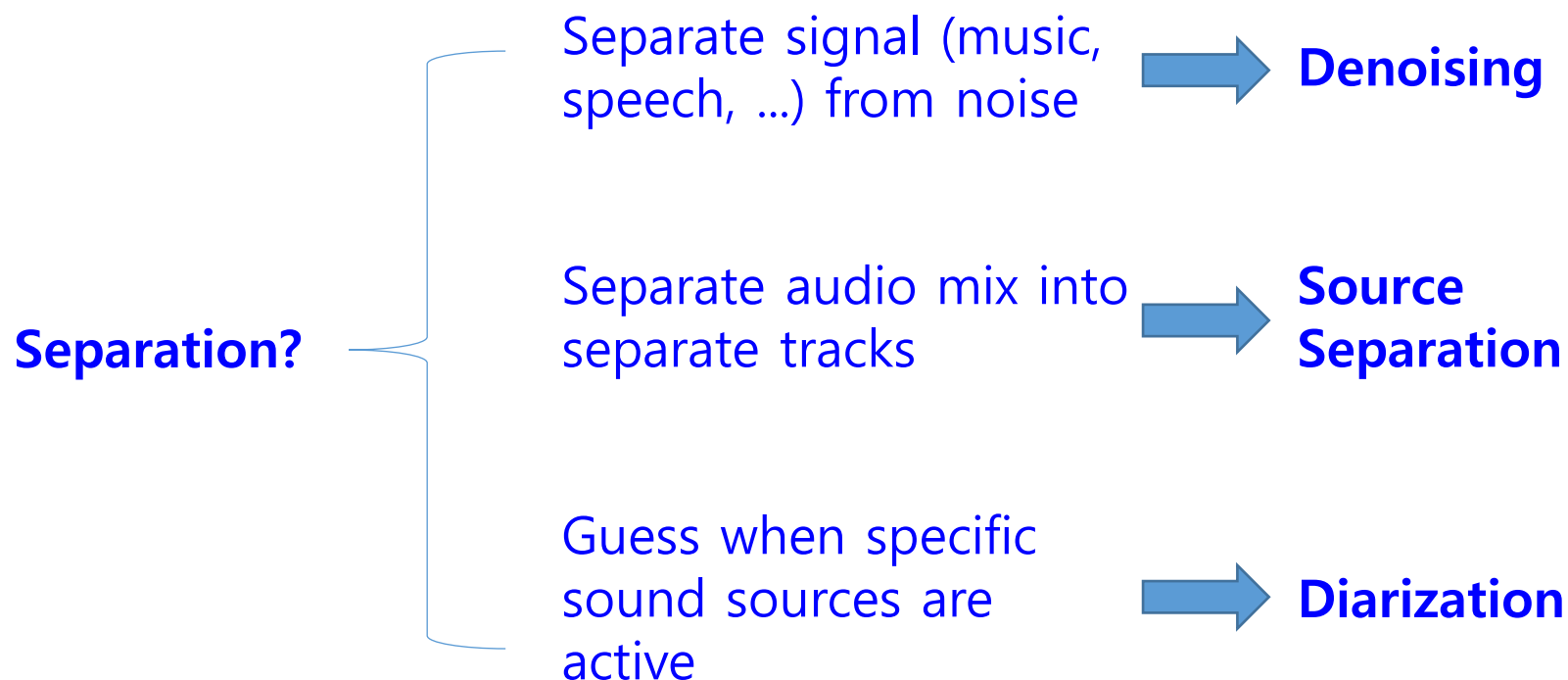


This lecture material refers to
https://github.com/yandexdataschool/speech_course?tab=readme-ov-file and
<https://github.com/markovka17/dla>



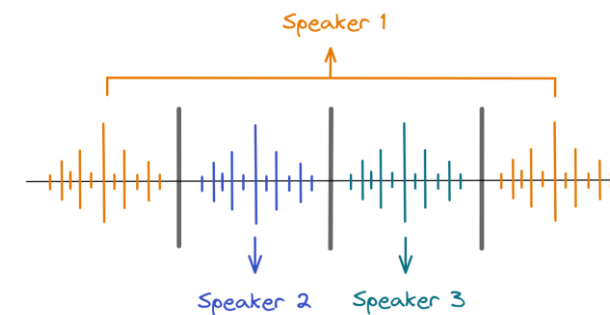
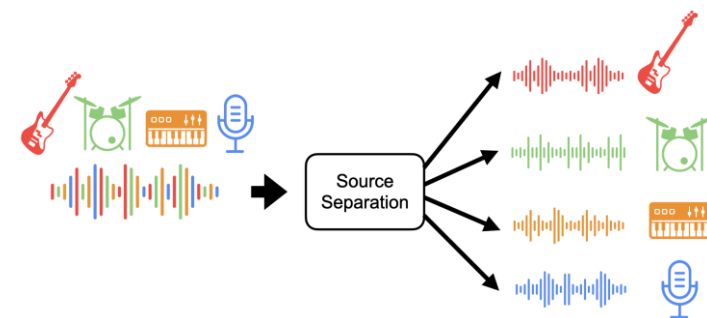
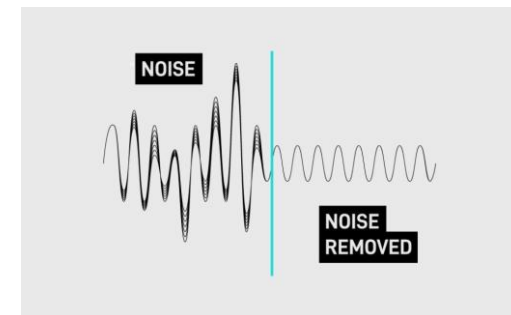
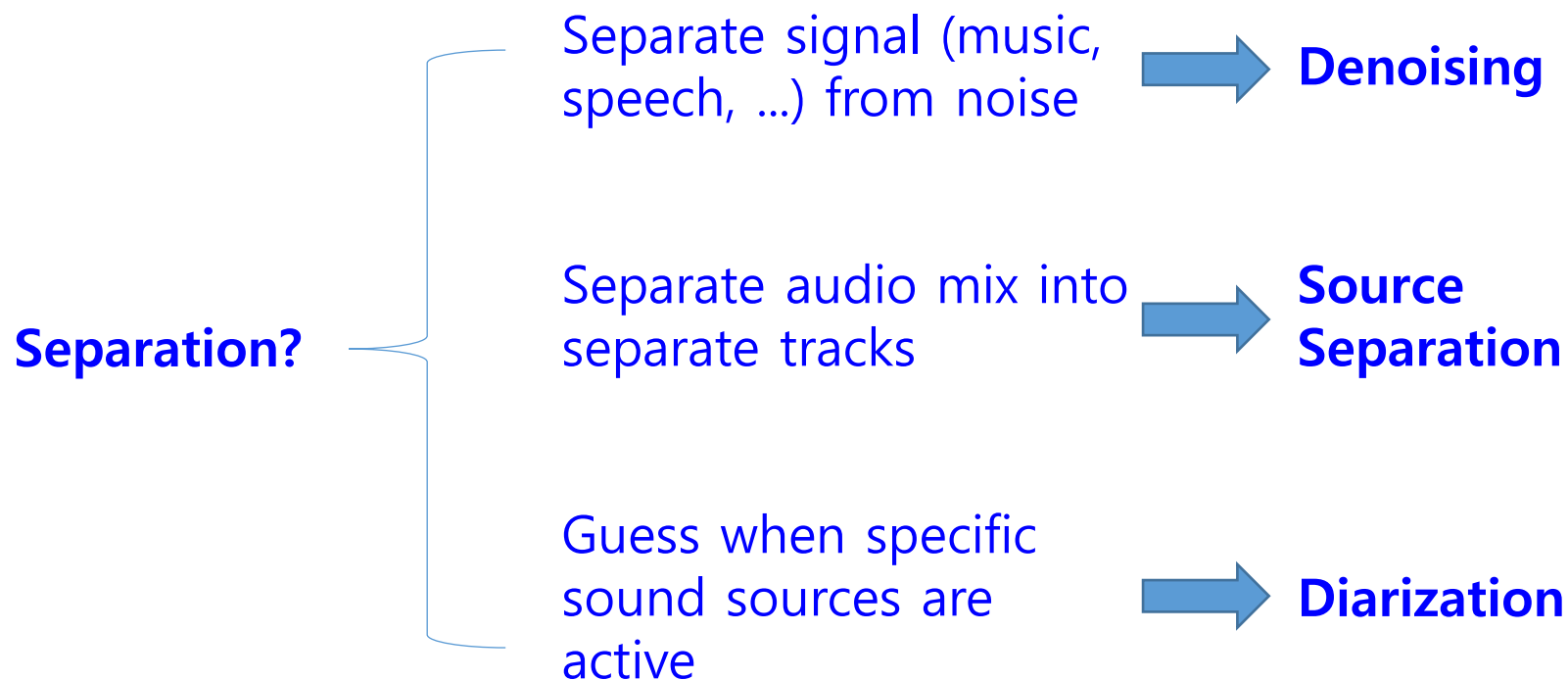
What is Source Separation?

- Source Separation literally means separating any source of particular interest...



What is Source Separation?

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What is Source Separation?

- Source Separation literally means separating any source of particular interest... **Source Separation + Denoising**

Separation?

Separate signal (music, speech, ...) from noise



Denoising

Separate audio mix into separate tracks

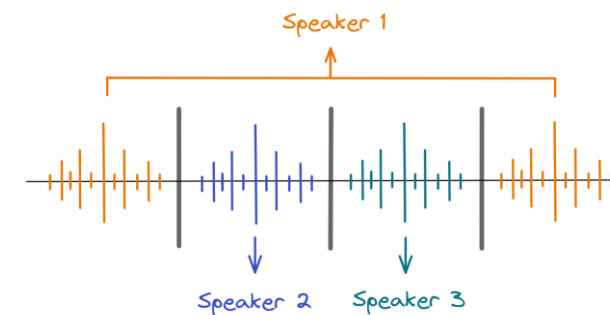
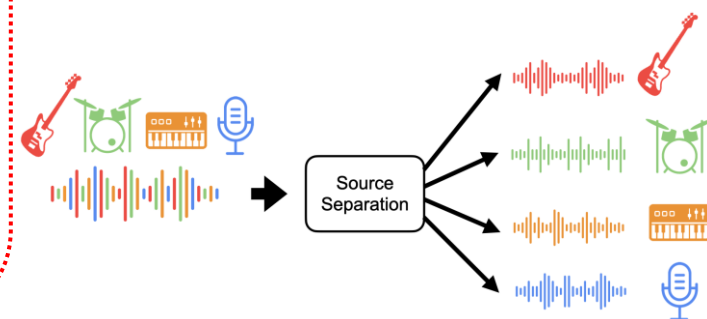
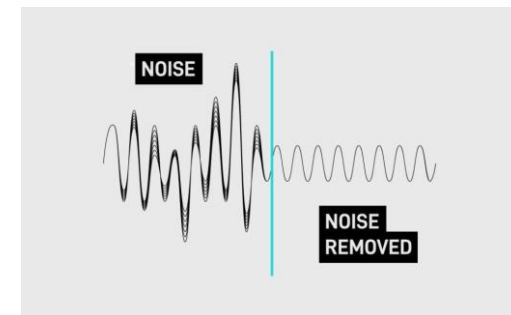


Source Separation

Guess when specific sound sources are active



Diarization



Denoising Problem

- **Goal:** guess what is noise and remove the noise from audio

Sound is Simple

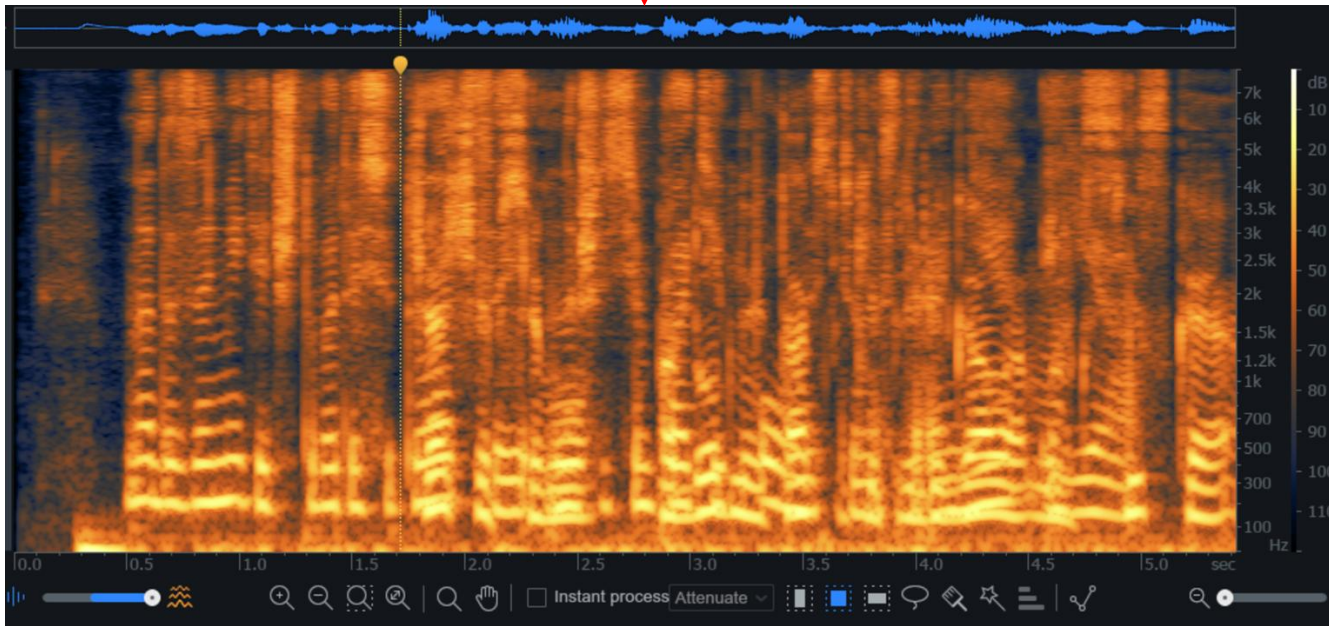


- **Spectrum-based Denoising**
 - **Main idea:** the noise is different, we will just cut it from the spectrum
 - **Early (and still popular) solution:** different types of spectral profiling

Denoising Problem

- **Goal:** guess what is noise and remove the noise from audio

Sound is complicated (Two people speaking)



- **Spectrum-based Denoising**
 - **Main idea:** the noise is different, we will just cut it from the spectrum
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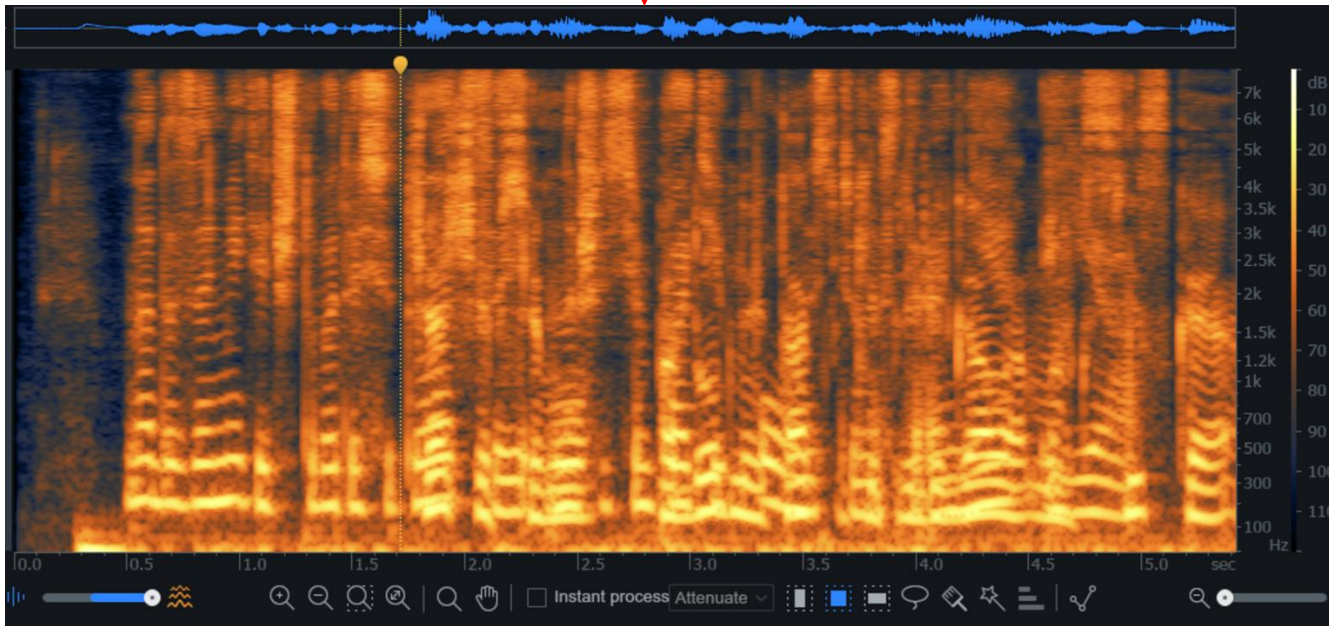
Denoising Problem

- **Goal:** guess what is noise and remove the noise from audio

Sound is complicated (Two people speaking)



The noise may not be well-separable

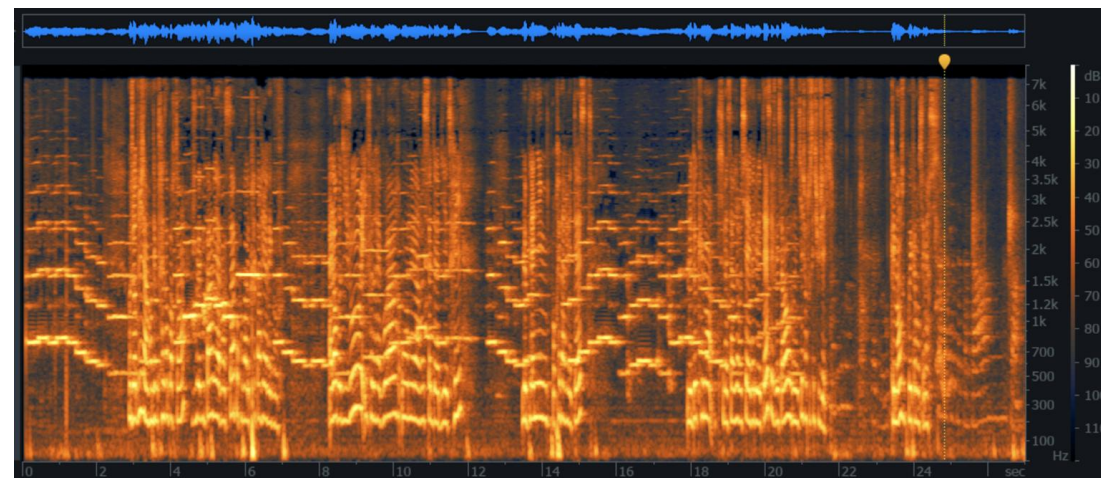


- **Spectrum-based Denoising**
 - **Main idea:** the noise is different, we will just cut it from the spectrum
 - **Early (and still popular) solution:** different types of spectral profiling

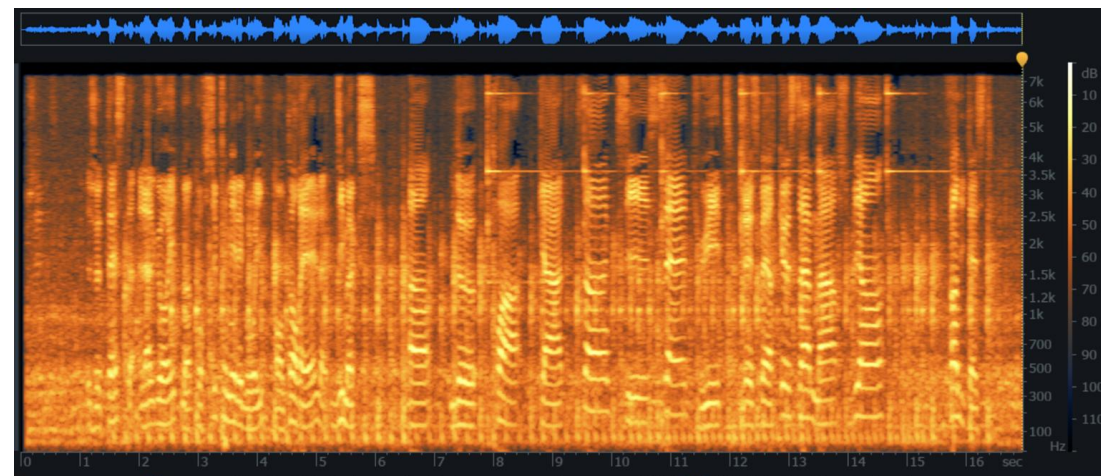
Denoising Problem

- **Deep learning approach:** see what can be done on the spectrogram
- **Main idea:** we'll still just cut it from the "spectrum"
- To separate complex audio, we need nontrivial ways

Sarah
&
Flute



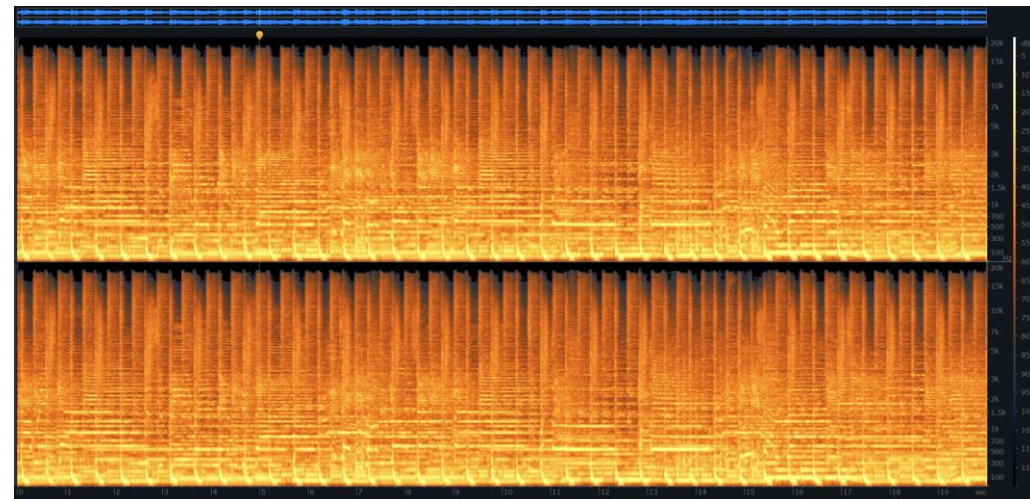
Alex
&
noise



Separation Problem

- **Deep learning approach:**
very similar

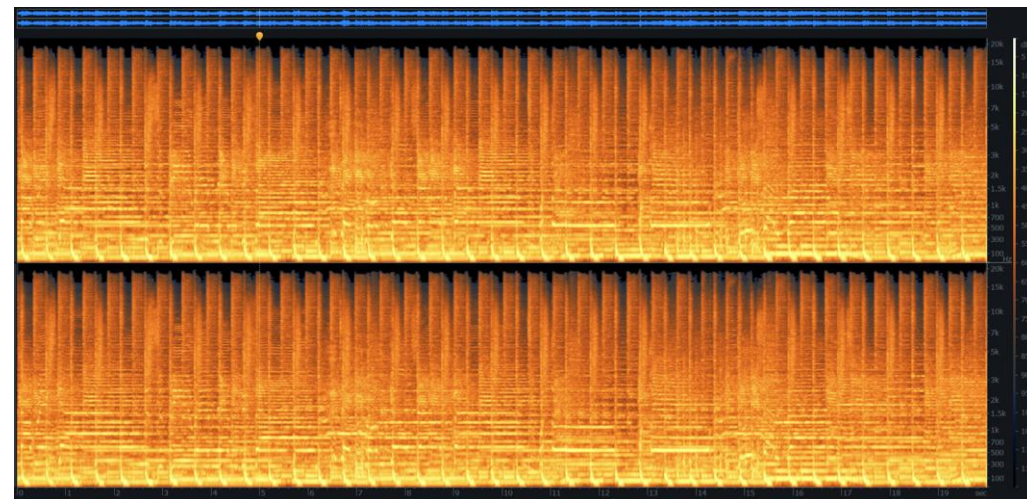
Song



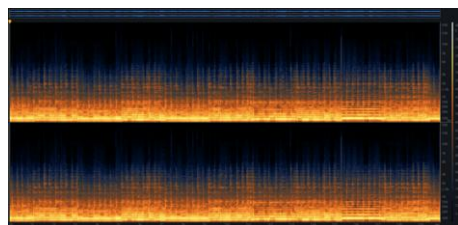
Separation Problem

- **Deep learning approach:** very similar
- **Main idea:** still just carve it from the "spectrum" (using DEMUCS)

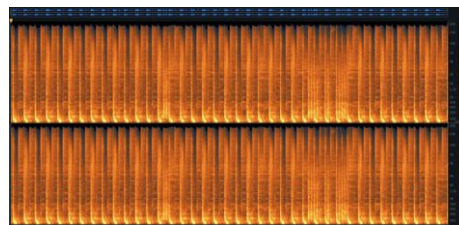
Song



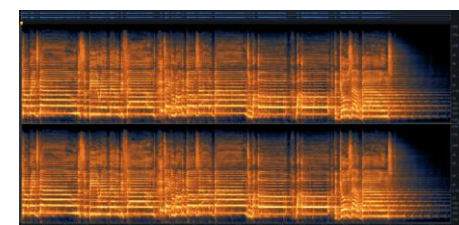
The Easton Ellises - Falcon 69



Bass:



Drums:



Vocals:



True:



True:



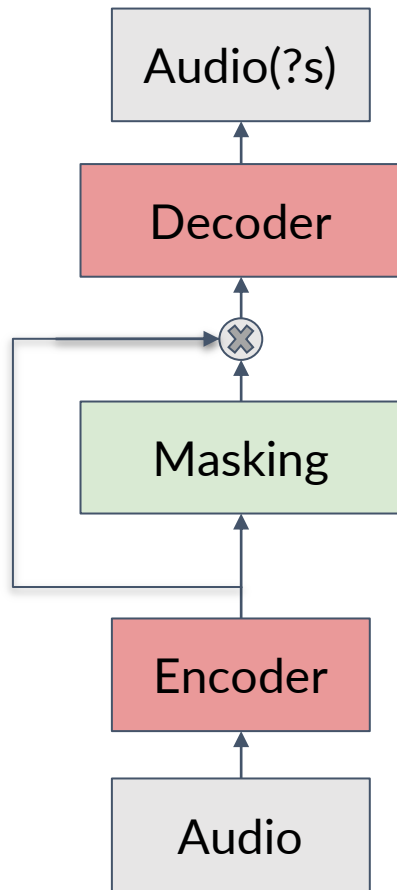
True:



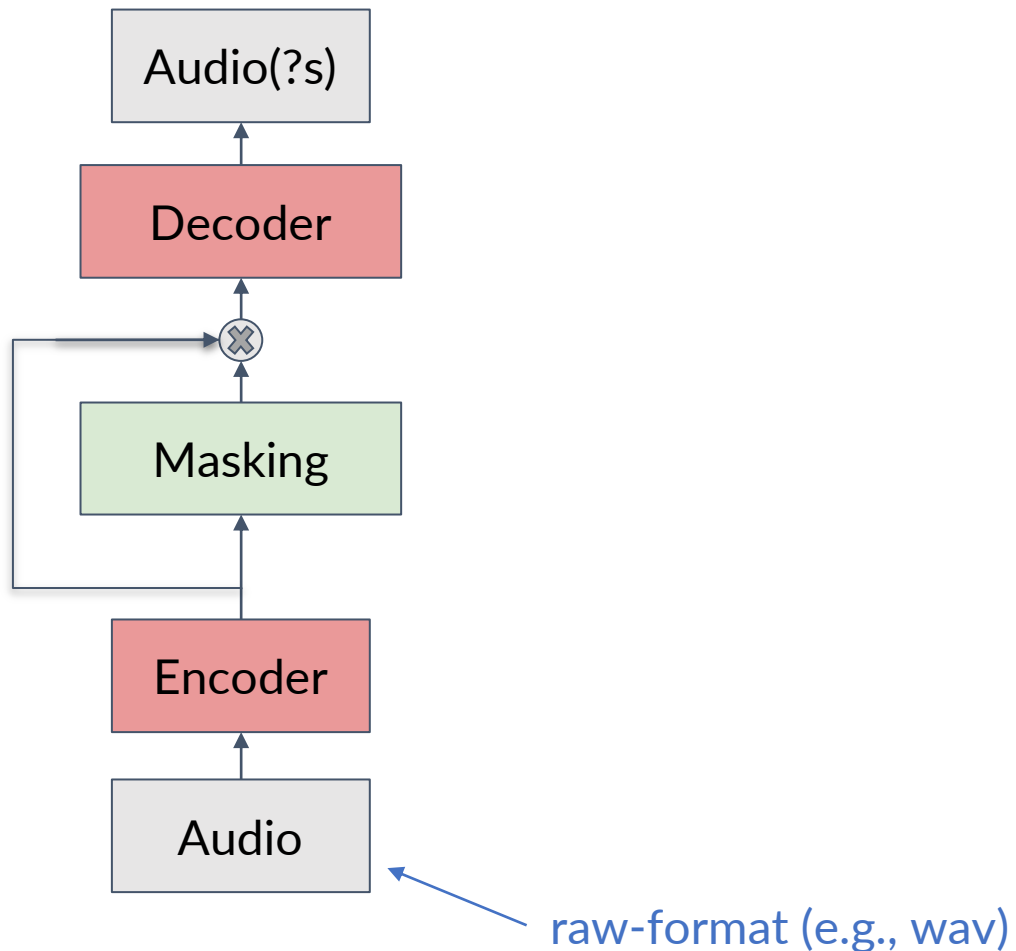
Separation Applications

- **Speech Enhancement (Denoising)**
 - 화상회의·통화 잡음 제거, 보청기/인공와우, ASR 전처리
- **Speech Separation (Cocktail-party problem)**
 - 다화자 회의록·ASR, Target Speaker Extraction
- **Music Source Separation**
 - 보컬/드럼/베이스 stem 추출, 카라오케·리믹스, 음악 정보 검색 (MIR)
- **Cinematic / Media Audio**
 - 대사·음악·효과음 (Dialogue/Music/Effects) 분리 → 리마스터링, 더빙
- **Robot Audition**
 - 로봇 음성 인터페이스, SELD·이상음 탐지 전처리 (다채널 분리와 결합)

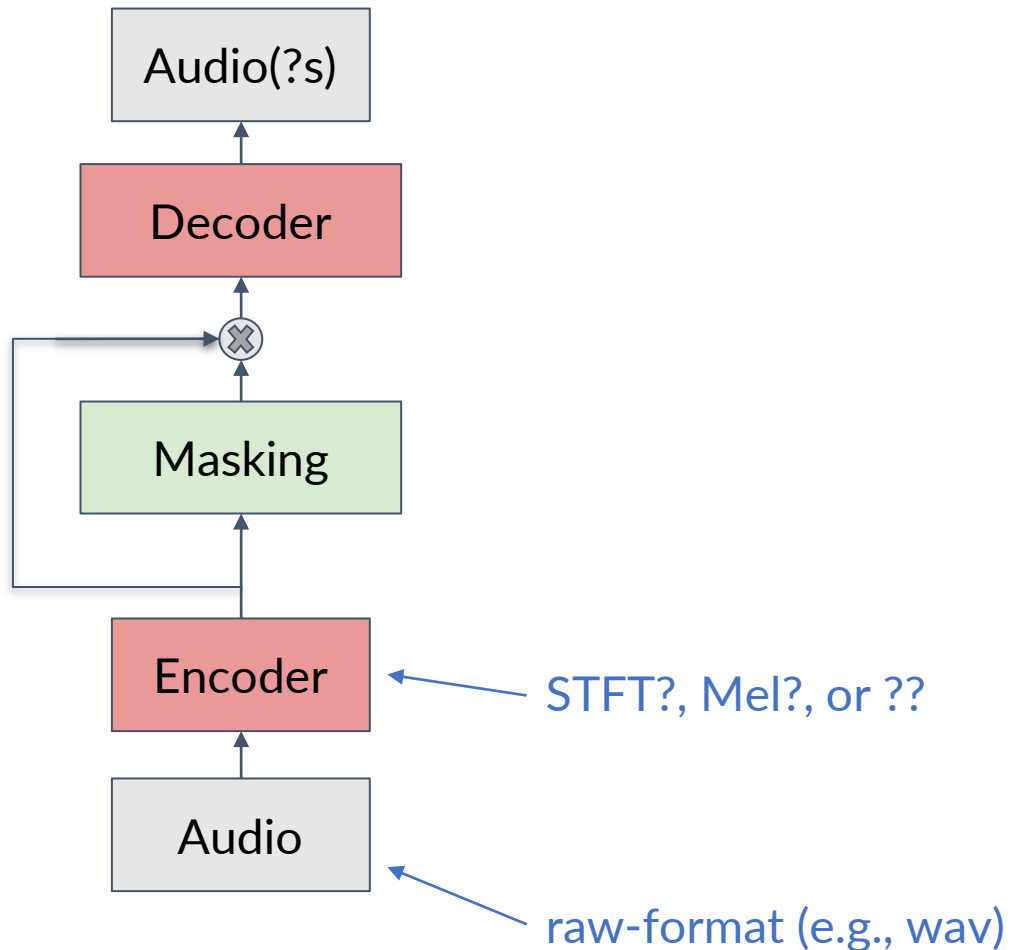
Denoising and Separation Pipeline



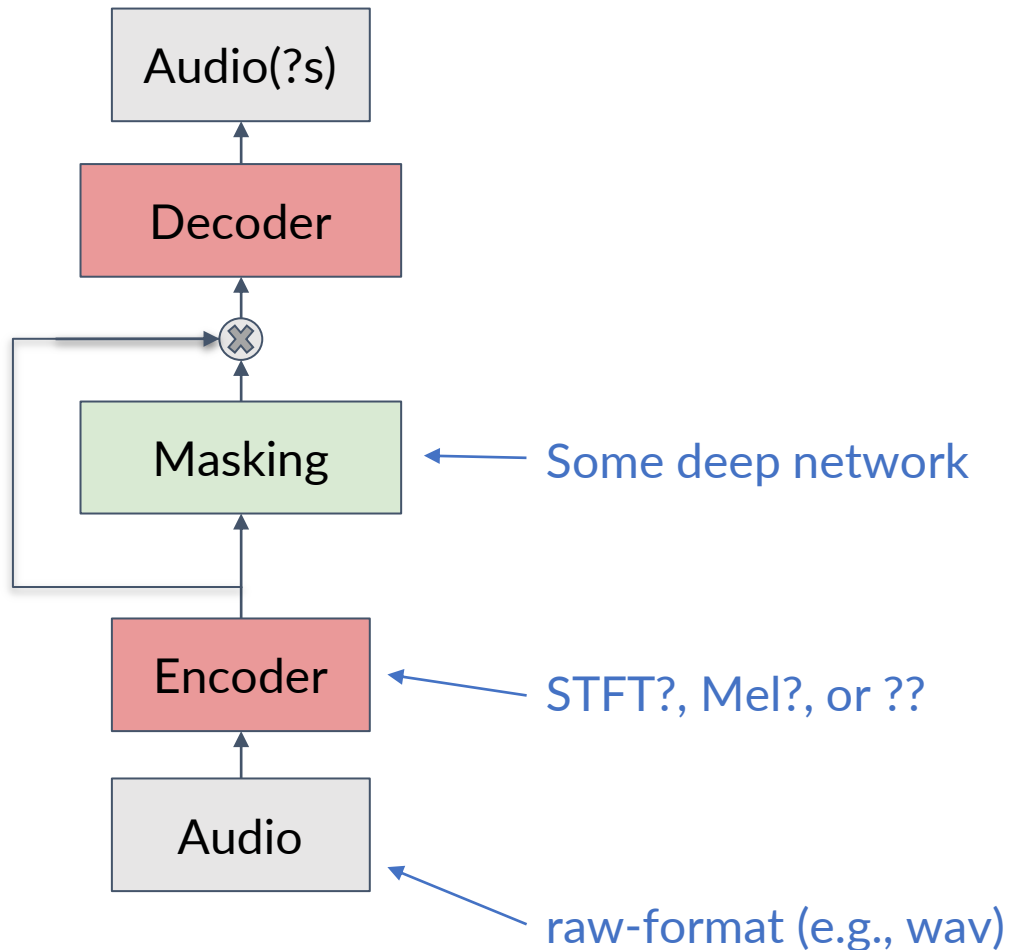
Denoising and Separation Pipeline



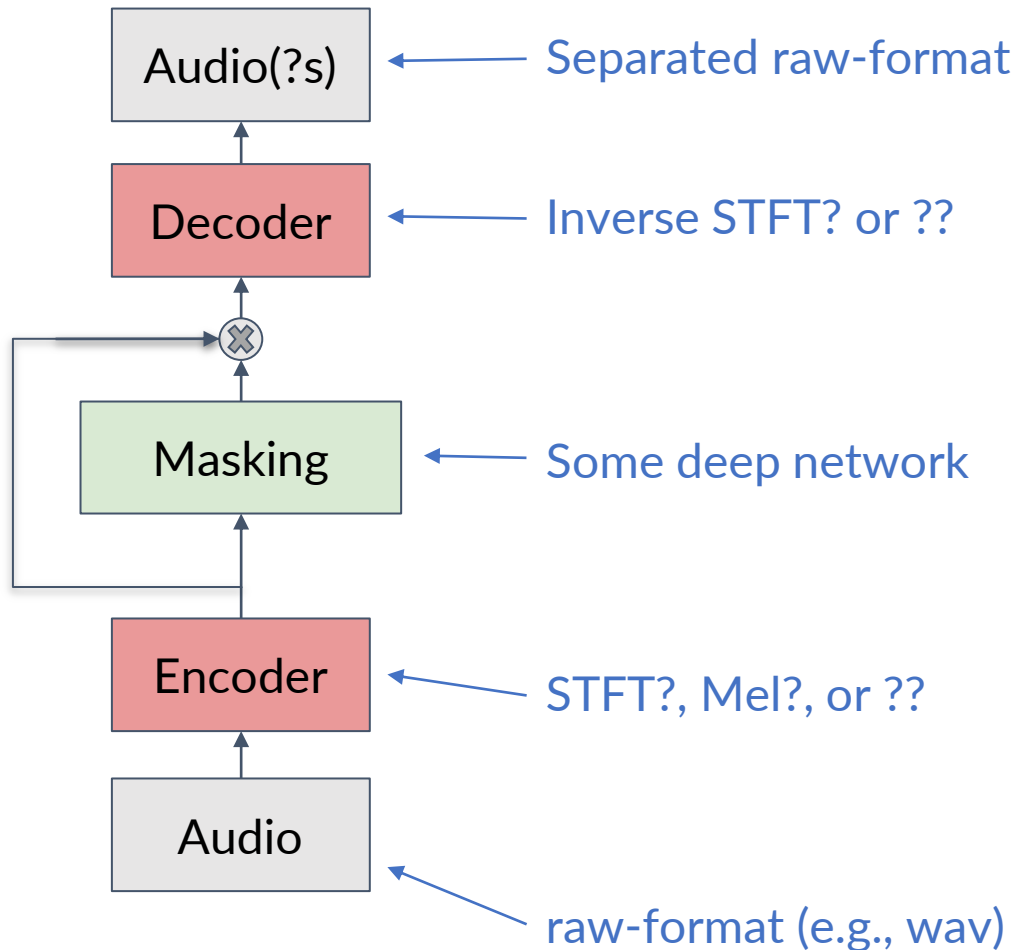
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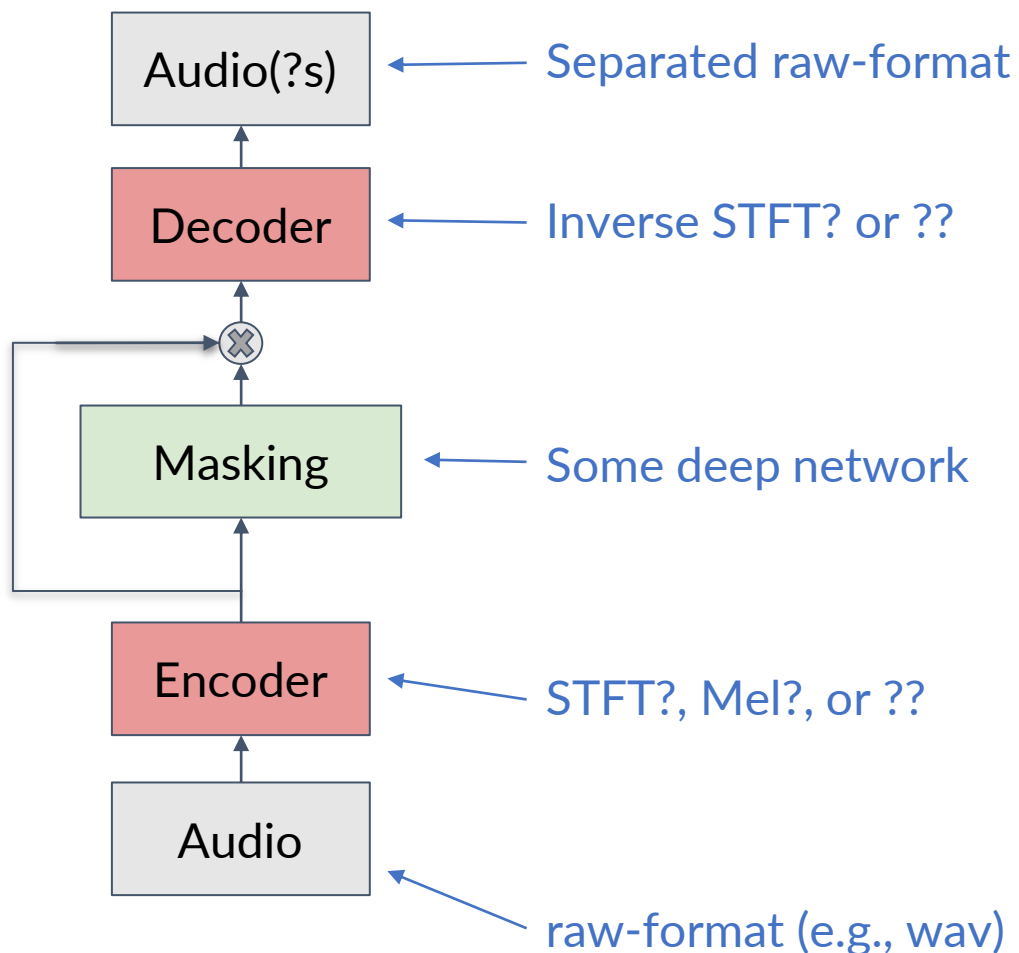
Denoising and Separation Pipeline



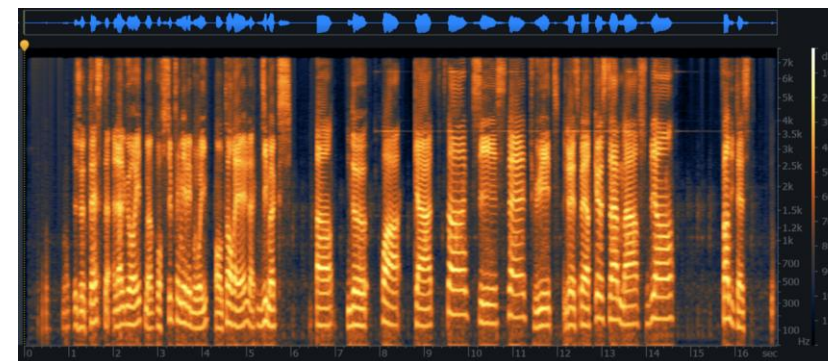
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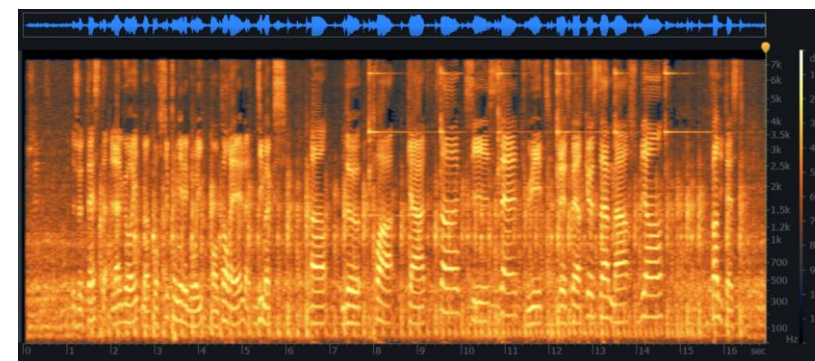
Denoising and Separation Pipeline



Denoised
Alex



Alex
&
noise

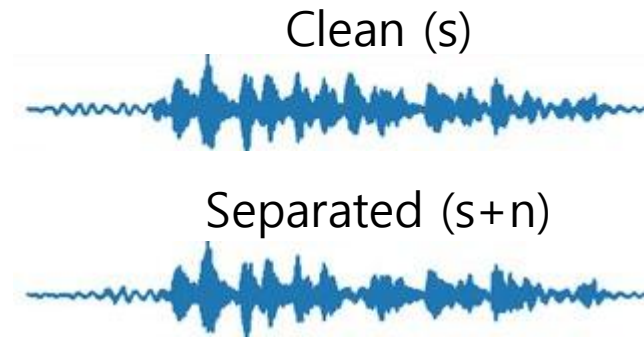
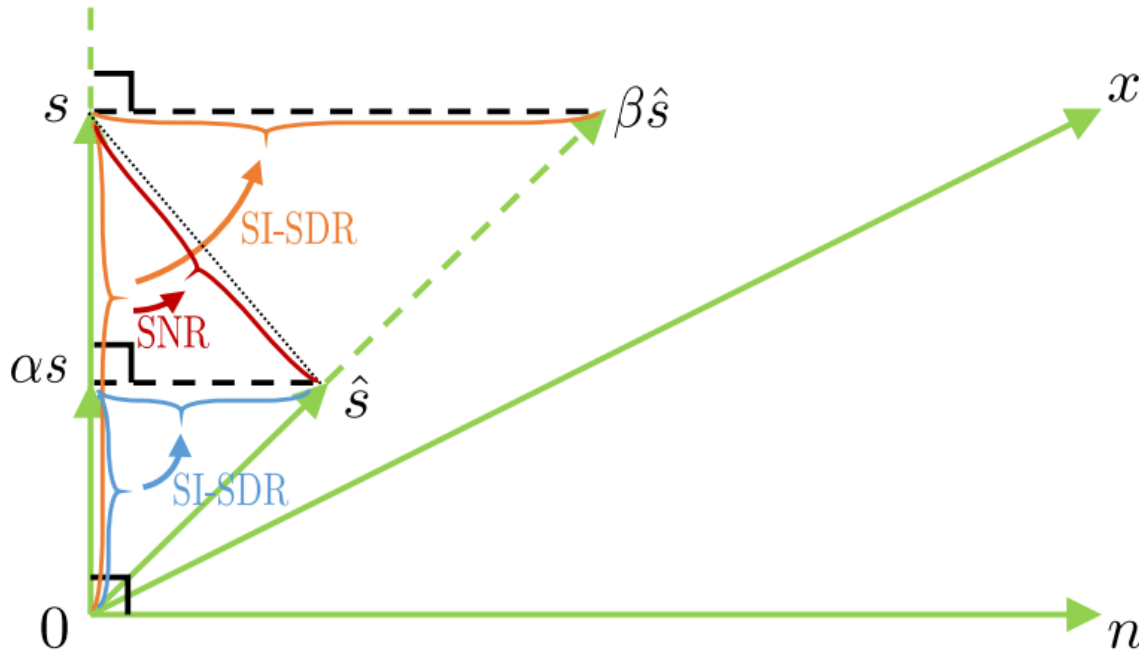


Denoising and Separation Metrics

- SNR (Signal-to-Noise Ratio) in dB:
- Si-SNR (Scale-invariant SNR) in dB:
- PESQ (Perceptual Evaluation of Speech Quality):
- STOI (Short-Time Objective Intelligibility):

Denoising and Separation Metrics

- SNR (Signal-to-Noise Ratio) and SI-SNR (Scale-invariant SNR)



$$SNR = 10 \log \frac{\|s\|^2}{\|s - \hat{s}\|^2}$$

$$SI_SNR = 10 \log \frac{\|\alpha s\|^2}{\|\alpha s - \hat{s}\|^2}$$

where $\alpha = \langle \hat{s}, s \rangle / \|s\|^2$ (s 를 $\hat{s}$ 방향으로 최적 스케일링)

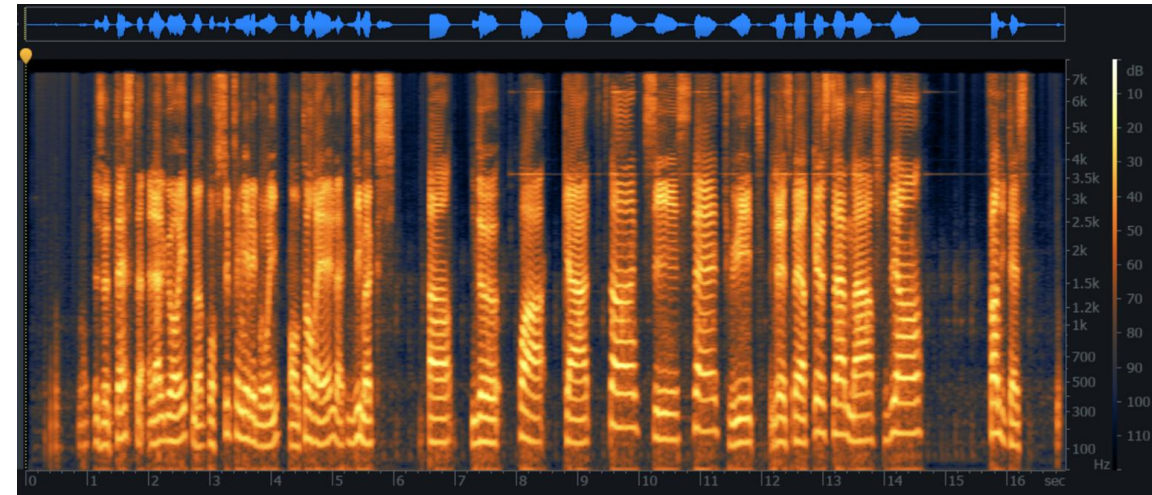
SI-SNR이 언제 효과적일까?

Denoising and Separation Metrics

- PESQ (Perceptual Evaluation of Speech Quality):
 - 사람이 느끼는 음질 (Perceptual quality)를 객관적으로 수치화하기 위한 지표 → 음질
 - -0.5 (bad)~4.5 (great)
- STOI (Short-Time Objective Intelligibility)
 - 사람이 얼마나 말을 알아들을 수 있는가 (intelligibility)를 예측하기 위한 지표 → 명로도
 - 0 (Bad)~1 (great)

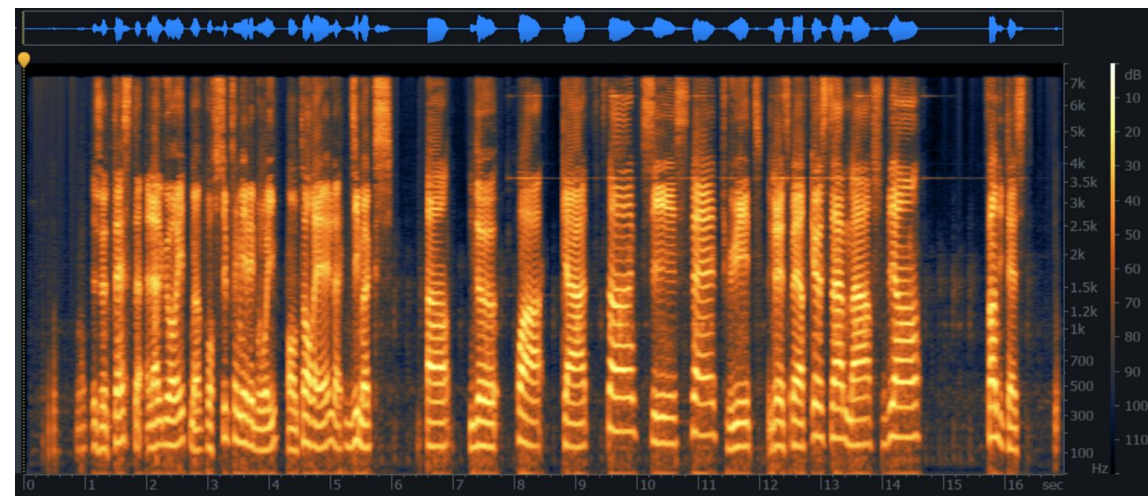
Spectrogram Information

- **Spectrogram** (Amplitude info.)
 - Convert the magnitude (or squared magnitude) of STFT to a dB scale

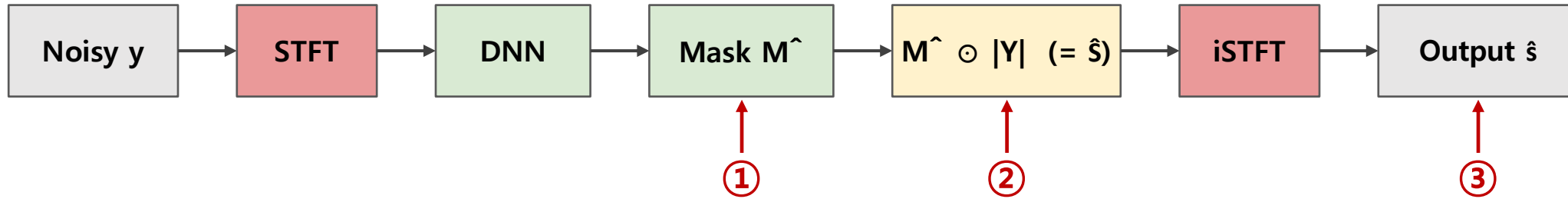


Spectrogram Information

- **Spectrogram** (Amplitude info.)
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 - **Idea:** use spectrogram as part of the encoder and inverse STFT as a decoder from masked spectrogram
- How to compute loss?



Loss Functions: Where to Compare?



① Mask Approximation (MA)

$$L = \|\hat{M} - M^*\|^2$$

- 정답 마스크(IRM, IBM 등)를 직접 맞힘
- 단순·안정적이지만, 마스크 오차 ≠ 소리 오차 ($|Y|$ 가 작은 칸의 오차도 똑같이 취급)

② Signal Approximation (SA)

$$L = \|\hat{M} \odot |Y| - |S|\|^2$$

- 마스크를 곱한 결과를 정답 크기 $|S|$ 와 비교
- 에너지가 큰 칸에 자동 가중치 → 보통 MA보다 우수 (Weninger et al., 2014)

③ Time-domain (Waveform)

$$L = \|\hat{s} - s\|_1 \text{ or } -\text{SI-SNR}(\hat{s}, s)$$

- iSTFT까지 통과한 최종 파형으로 비교 (end-to-end)
- 위상 오차까지 loss에 자동 반영

출력에 가까운 곳에서 비교할수록 실제 품질 지표(SI-SNR, PESQ)와 잘 맞음 ↔ 대신 iSTFT/Decoder까지 미분 가능하게 연결해야 함

Spectral Losses & Phase

Phase-sensitive SA (PSA)

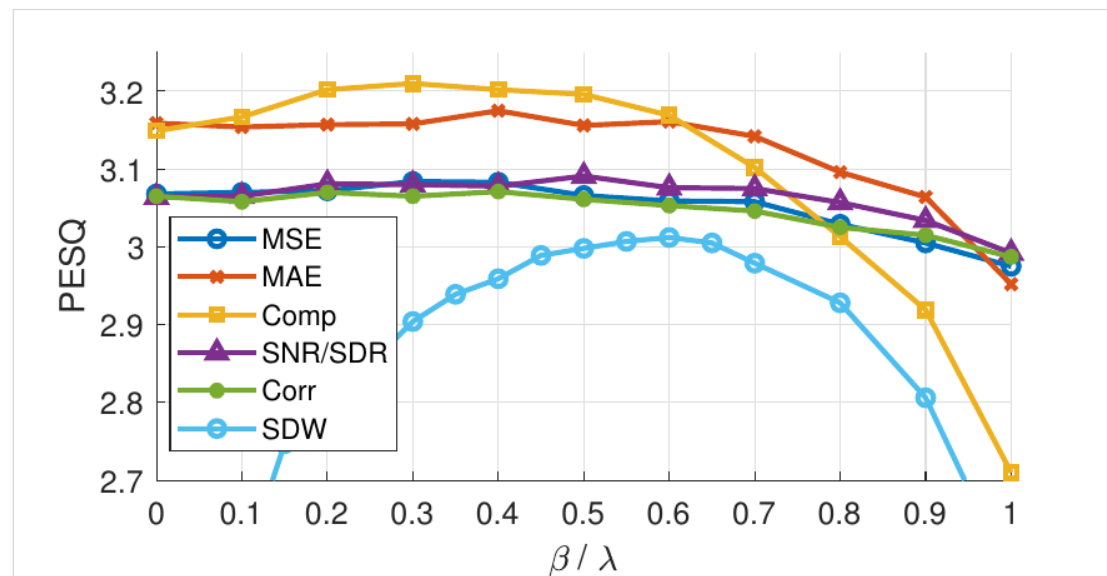
- 목표를 $|S| \cdot \cos(\theta_S - \theta_Y)$ 로 바꿈 → 위상이 어긋난 칸은 목표 크기를 줄임
- $L = \|\hat{M} \odot |Y| - |S| \cos(\theta_s - \theta_y)\|^2$ (Erdogan et al., 2015)

Power-law compressed loss ($c \approx 0.3$)

- 작은 성분도 학습에 반영 (로그처럼 압축, 사람의 음량 지각과 유사)
- $L_{mag} = \sum (|s|^c - |\hat{s}|^c)^2$

Magnitude + Complex 혼합

- $L = (1 - \beta) \cdot L_{mag} + \beta \cdot L_{complex}$
- $L_{complex} = \sum \left| |S|^c e^{j\varphi_S} - |\hat{S}|^c e^{j\varphi_{\hat{S}}} \right|^2$ (위상까지 비교)
- $\beta = 0$: 크기만, $\beta = 1$: 복소(위상 포함)만

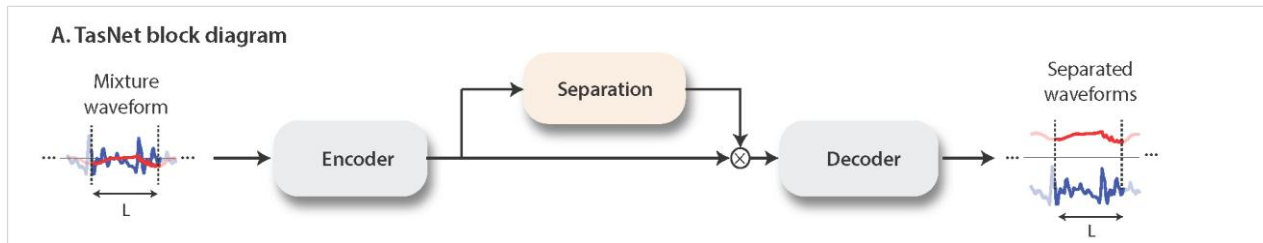


PESQ vs. 복소 loss 가중치 β . 크기와 복소 loss를 섞을 때 모든 거리 척도에서 항상, compressed loss는 $\beta \approx 0.3$ 에서 최고

Braun & Tashev, "A consolidated view of loss functions for supervised deep learning-based speech enhancement," 2020 (arXiv:2009.12286)

크기만 맞는 loss → 위상 정보를 조금씩 넣는 loss로 발전

Time-domain Loss: SI-SNR & PIT



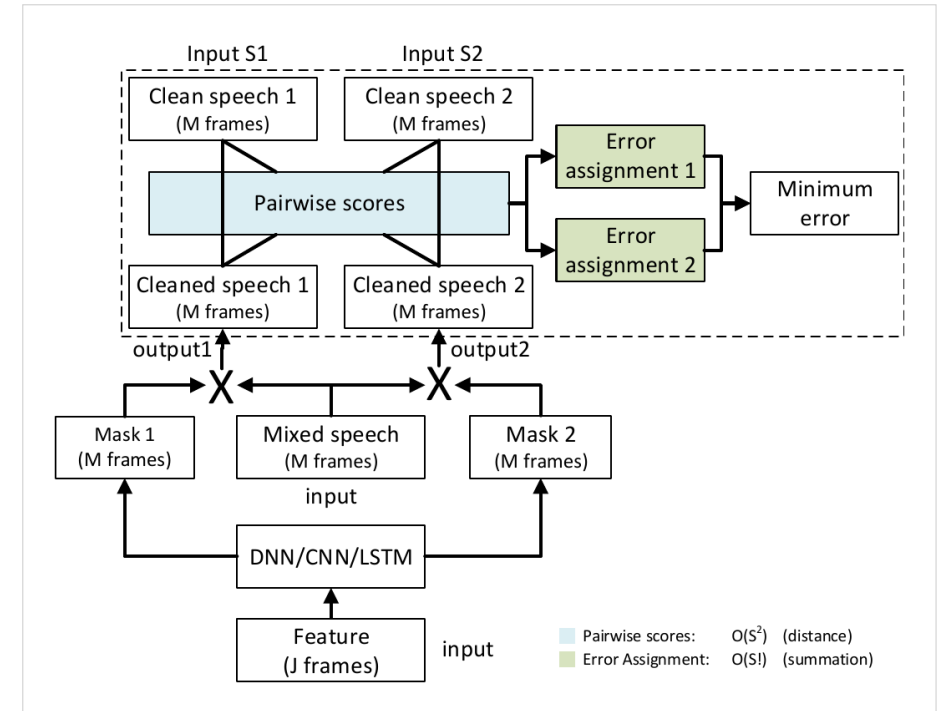
Conv-TasNet (Luo & Mesgarani, 2019, arXiv:1809.07454): 파형 $\rightarrow$ Encoder $\rightarrow$ Mask $\rightarrow$ Decoder, SI-SNR 로 end-to-end 학습

SI-SNR loss: $L = -\text{SI-SNR}(\hat{s}, s)$

- 19장의 평가 지표를 그대로 loss로 사용
- 크기 불변 $\rightarrow$ 출력 음량이 아니라 파형 모양에 집중
- Decoder(iSTFT) 이후 파형으로 계산 $\rightarrow$ 위상까지 자동 학습

다른 파형 기반 loss

- DEMUCS Denoiser: L1(파형) + Multi-resolution STFT loss
- 실전에서는 파형 loss + 스펙트럼 loss를 섞어 쓰는 경우가 많음



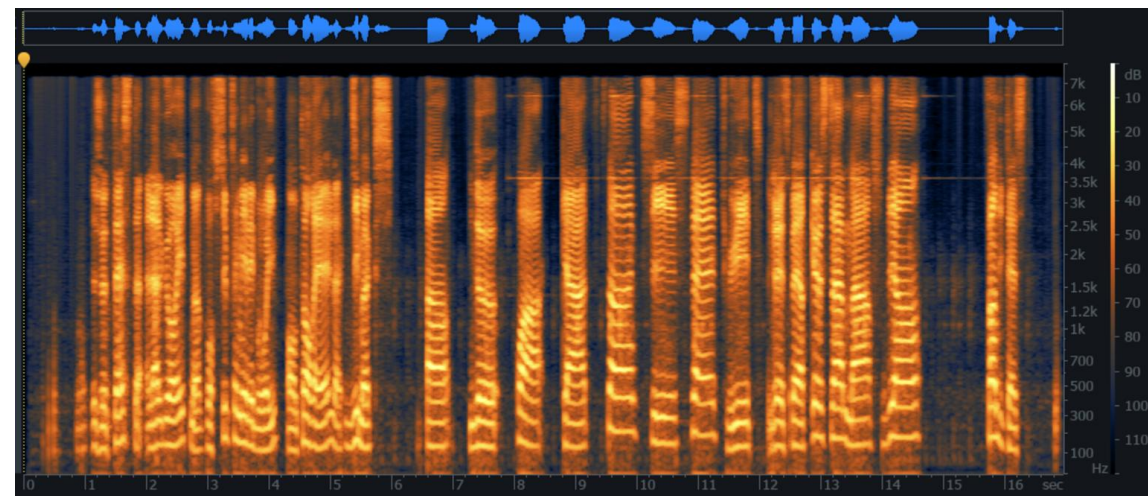
PIT (Permutation Invariant Training)

출력과 정답을 모든 순서 조합으로 비교해 가장 작은 오차를 loss로 사용: $L = \min_{\pi} \sum_i L(s_i, \hat{s}_{\pi(i)})$

Kolbæk et al., 2017 (arXiv:1703.06284)

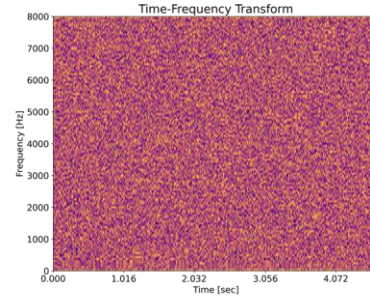
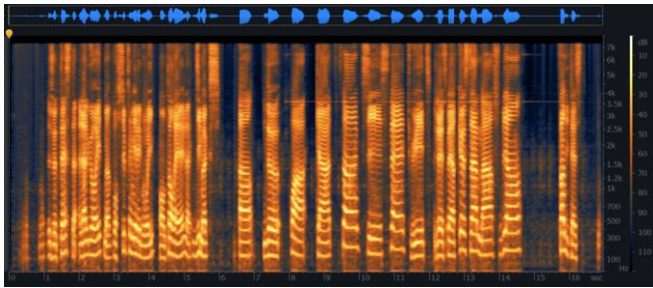
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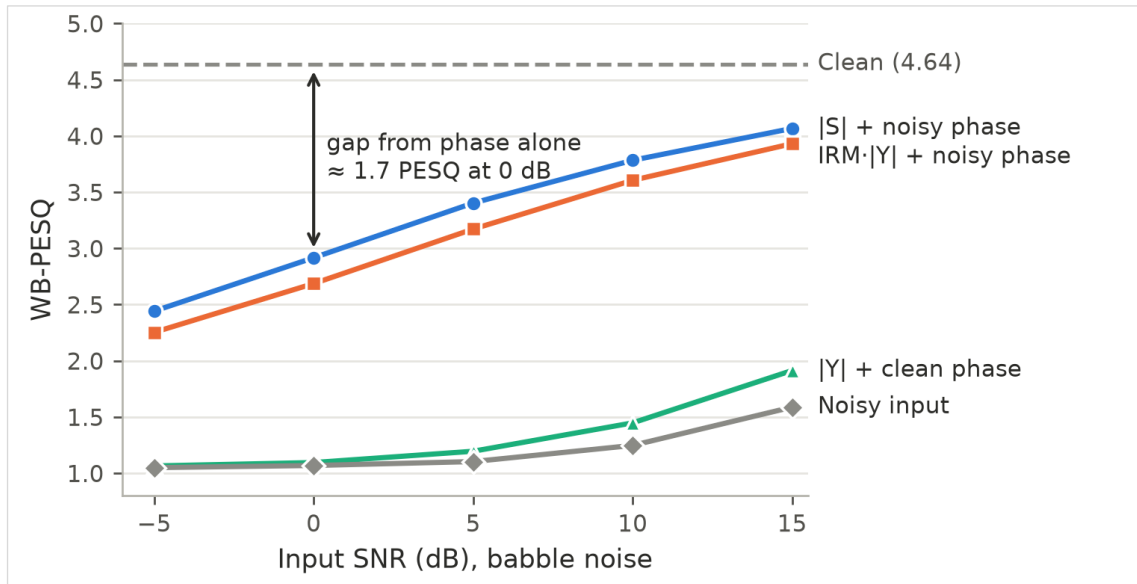


Oracle 실험: 위상은 얼마나 중요한가?

LibriSpeech 3문장 + babble-white 잡음, 입력 SNR $-5 \sim 15$ dB

- 정답 크기 $|S|$ 를 알아도 잡음 위상 φ_Y 를 쓰면 0 dB에서 PESQ 2.92 (정답 4.64)
- 위상만 정답 ($|Y| + \varphi_S$)이면 PESQ는 거의 그대로 $\rightarrow$ 크기가 먼저, 그다음 병목은 위상

How can we use phase effectively?

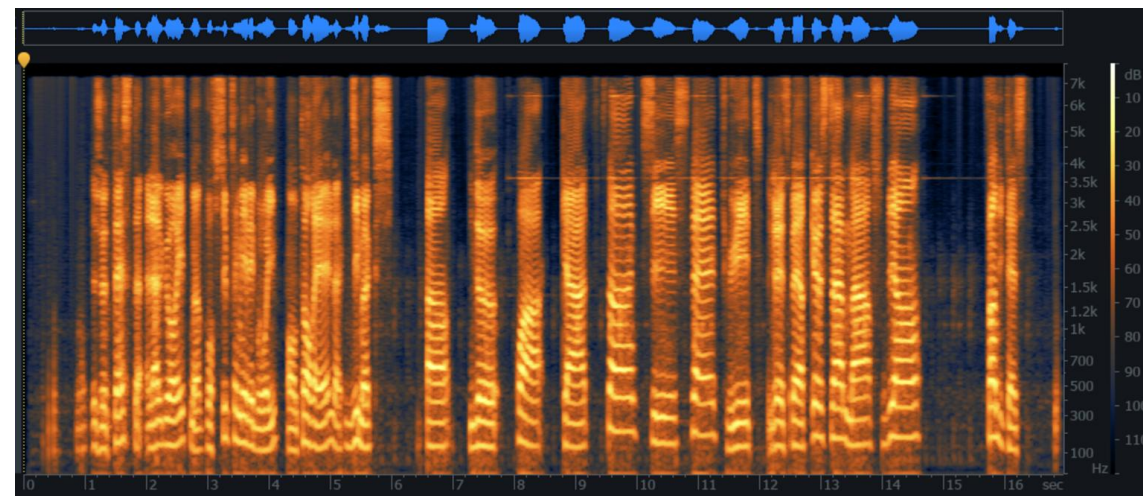


0 dB babble 잡음, 3문장 평균 (STFT 512, hop 128)

조건	PESQ	STOI	SI-SDR (dB)
Clean (상한)	4.64	1.00	∞
Noisy input Y	1.07	0.63	0.0
$IRM \cdot Y $ + 잡음 위상	2.69	0.94	9.5
S + 잡음 위상 φ_Y	2.92	0.97	10.2
$ S $ + Griffin-Lim 위상	4.36	1.00	-18.6
$ S $ + 무작위 위상	1.25	0.85	-40.5
$ Y $ + 정답 위상 φ_S	1.10	0.71	4.1

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- How to handle Phase info.?
- How to encode Spectrogram?

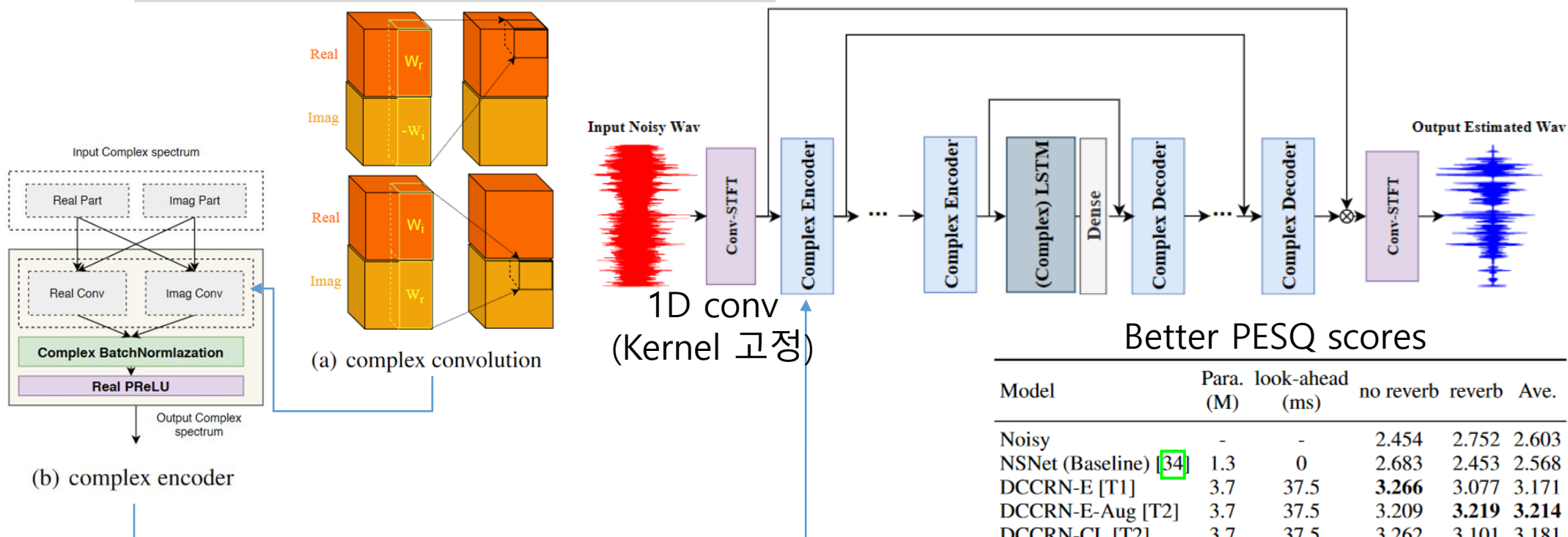


DCCRN (DNS Challenge, 2020)

- **DCCRN (2020)**

- Why not use the STFT coefficient (with phases) directly?
- Complex operations with complex data

$$\begin{aligned} \text{out}_r &= \text{conv}_r(x_r) - \text{conv}_i(x_i) & \# \text{ 실수부} \\ \text{out}_i &= \text{conv}_i(x_r) + \text{conv}_r(x_i) & \# \text{ 허수부} \end{aligned}$$



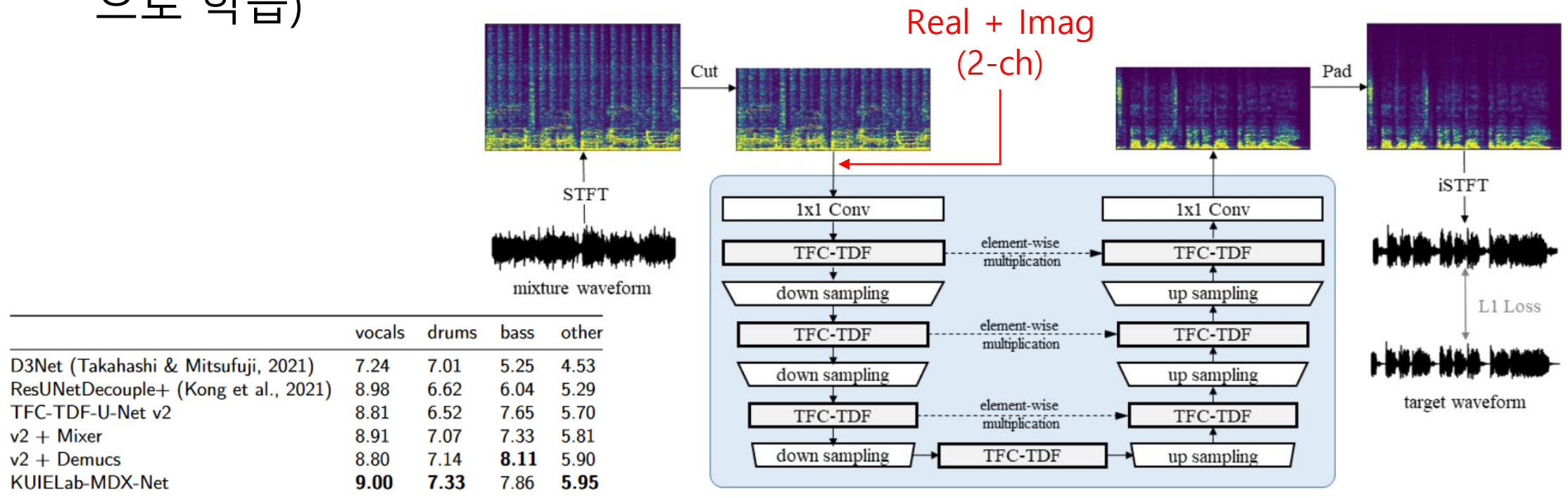
Better PESQ scores

Model	Para. (M)	look-ahead (ms)	no reverb	reverb	Ave.
Noisy	-	-	2.454	2.752	2.603
NSNet (Baseline) [34]	1.3	0	2.683	2.453	2.568
DCCRN-E [T1]	3.7	37.5	3.266	3.077	3.171
DCCRN-E-Aug [T2]	3.7	37.5	3.209	3.219	3.214
DCCRN-CL [T2]	3.7	37.5	3.262	3.101	3.181
DCUNET [T2]	3.6	37.5	3.223	2.796	3.001

MDX-Net (2021)

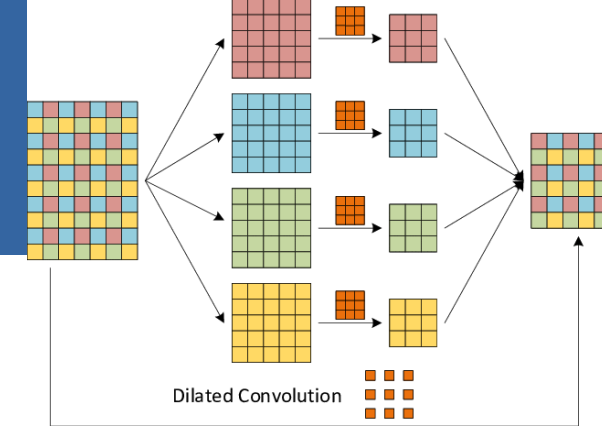
- **MDX-Net (2021)**

- TFC-TDF blocks (Time-Frequency Conv + Time-Distributed FC) in a U-Net structure
- Separation model for each category + Mixer (즉, 각 소스별로 독립적으로 학습)



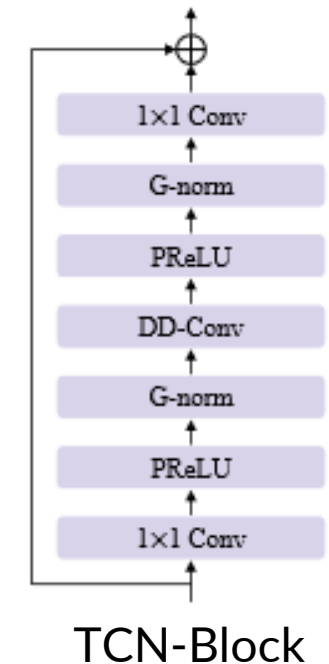
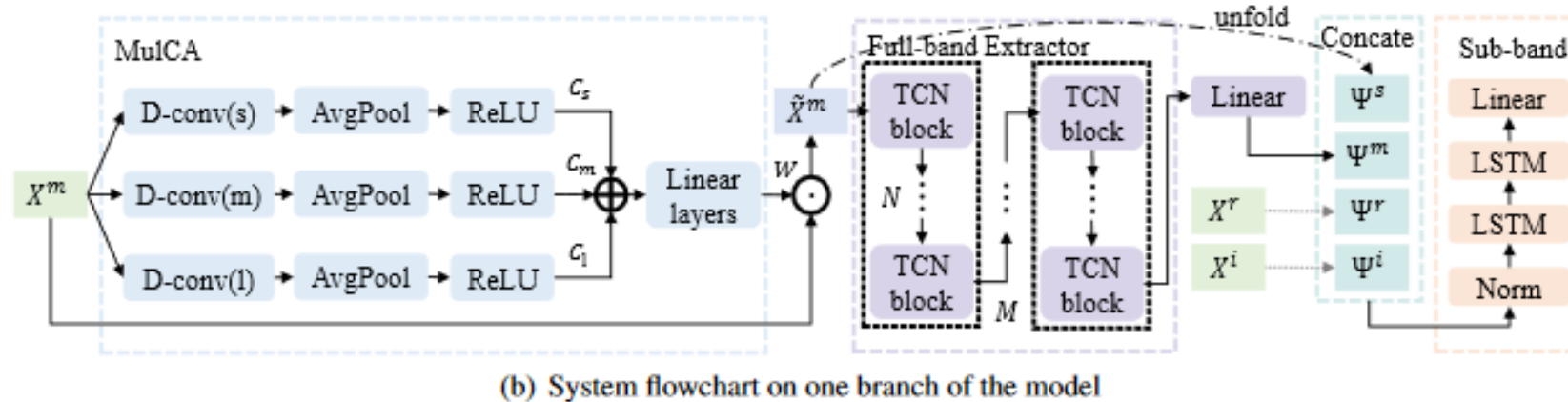
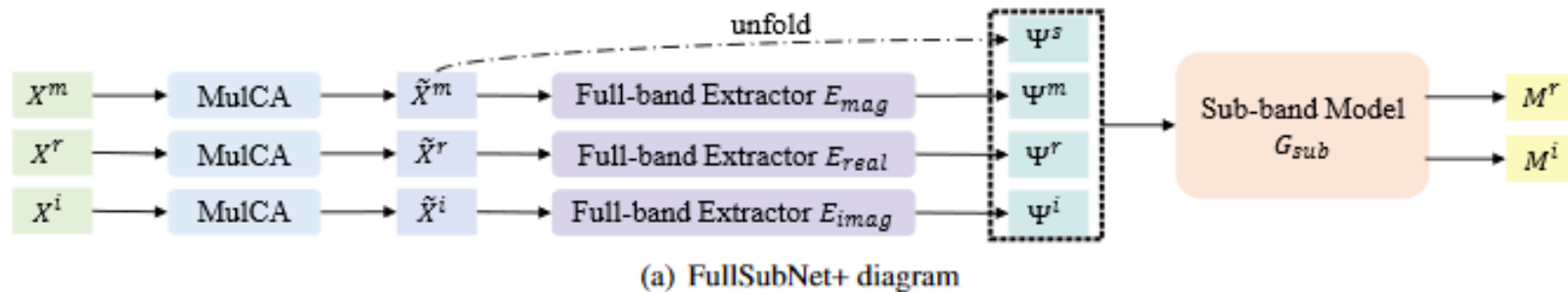
SDR (dB) on MUSDB18

FullSubNet+ (2022)



- **FullSubNet+ (2022)**

- **Idea:** magnitude + real/imag spectrograms $\rightarrow$ MulCA (multi-scale dilated conv) $\rightarrow$ TCN full-band + LSTM sub-band



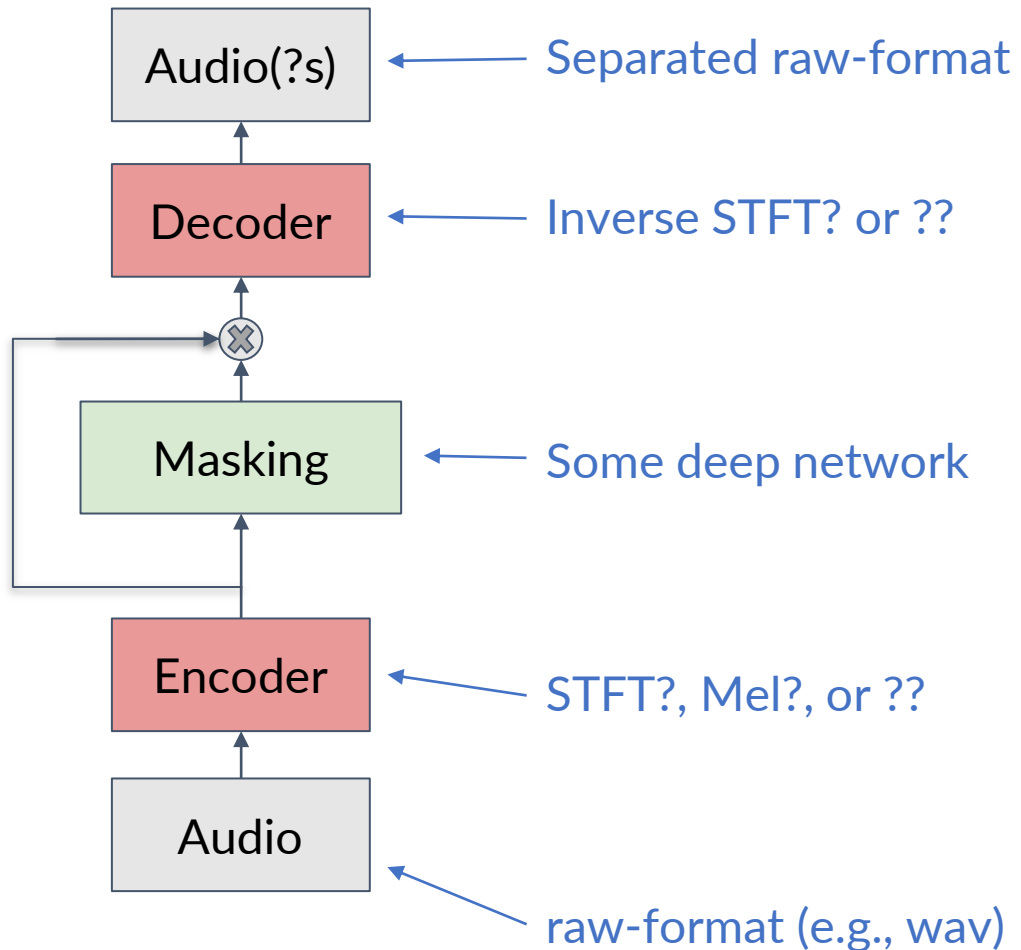
FullSubNet+(2022)

- **FullSubNet+(2022)**
 - **Idea:** use separately magnitude and 2-component phase, encode it via dilated convolutions then fully convolutional

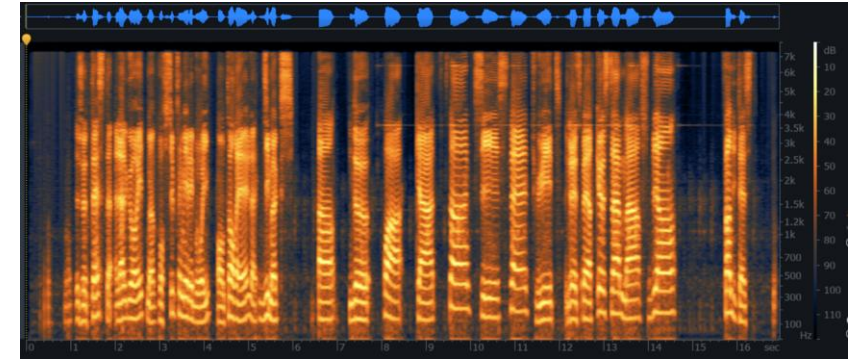
Table 1. The performance in terms of WB-PESQ [MOS], NB-PESQ [MOS], STOI [%], and SI-SDR [dB] on the DNS Challenge test dataset.

Model	Year	Look Ahead (ms)	With Reverb				Without Reverb			
			WB-PESQ	NB-PESQ	STOI	SI-SDR	WB-PESQ	NB-PESQ	STOI	SI-SDR
Noisy	-	-	1.822	2.753	86.62	9.033	1.582	2.454	91.52	9.07
DCCRN-E [22]	2020	37.5	-	3.077	-	-	-	3.266	-	-
PoCoNet [23]	2020	-	2.832	-	-	-	2.748	-	-	-
DCCRN+ [24]	2021	10	-	3.300	-	-	-	3.330	-	-
TRU-Net [25]	2021	0	2.740	3.350	91.29	14.87	2.860	3.360	96.32	17.55
CTS-Net [26]	2021	-	3.020	3.470	92.70	15.58	2.940	3.420	96.66	17.99
FullSubNet [12]	2021	32	3.063	3.581	92.93	16.09	2.813	3.403	96.17	17.44
FullSubNet+	2021	32	3.218	3.666	93.84	16.81	2.982	3.504	96.69	18.34

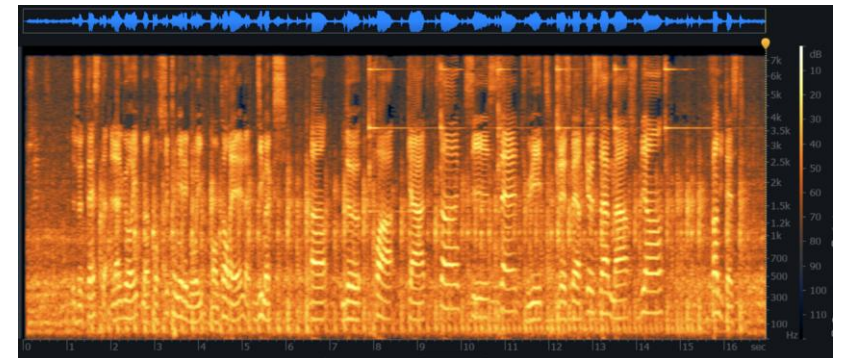
DEMUCS Denoiser (2020)



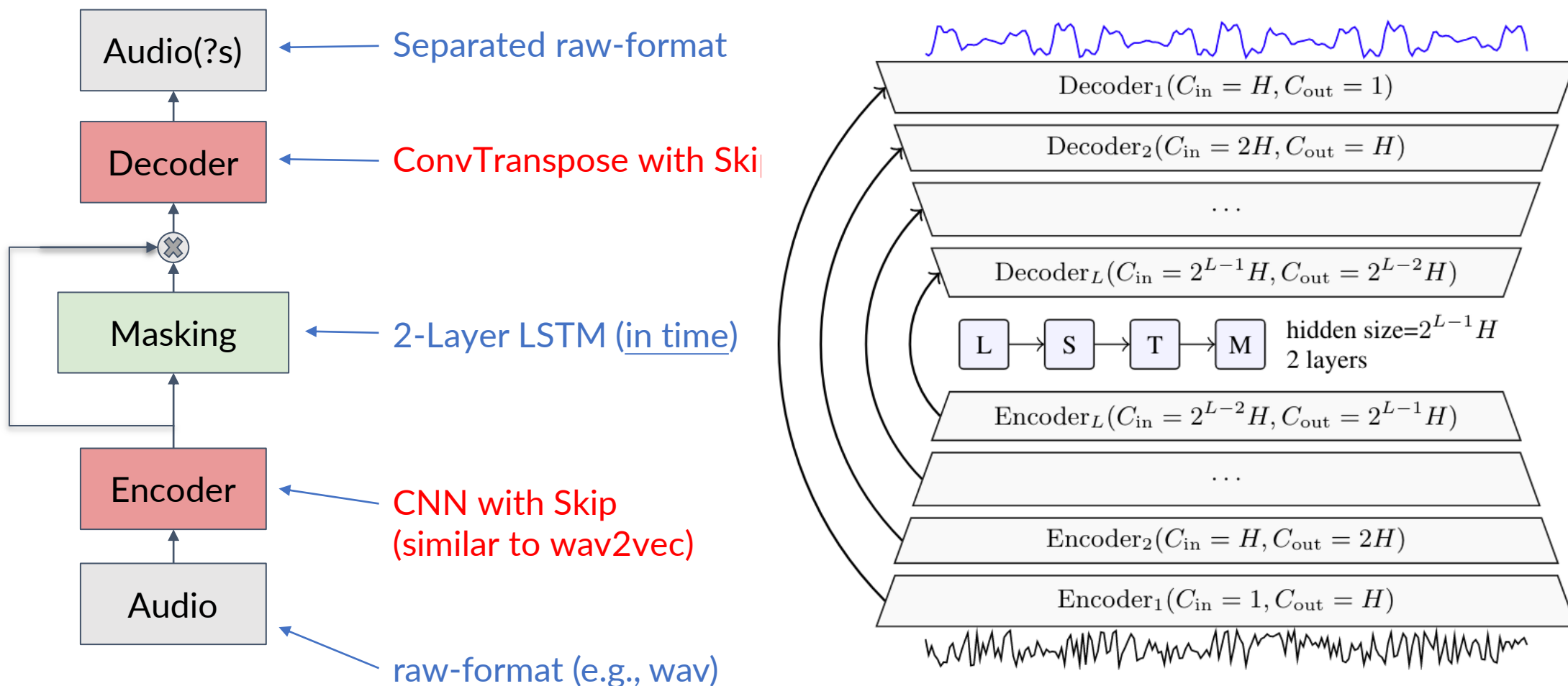
Denoised
Alex



Alex
&
noise



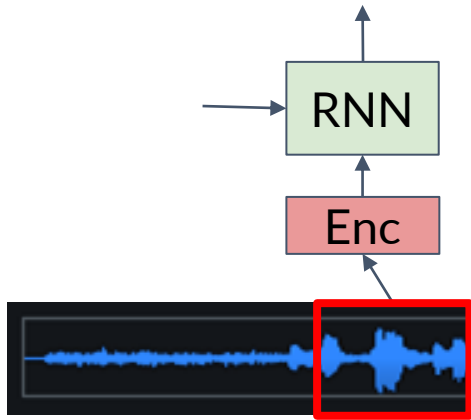
DEMUCS Denoiser (2020)



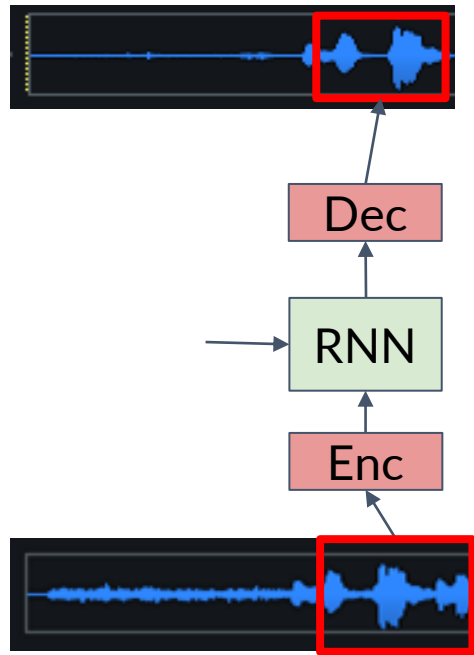
DEMUCS Denoiser (2020)



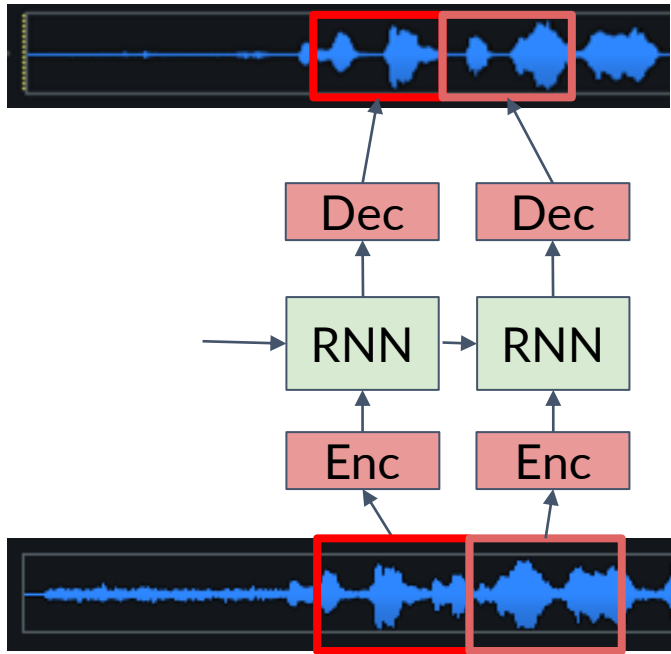
DEMUCS Denoiser (2020)



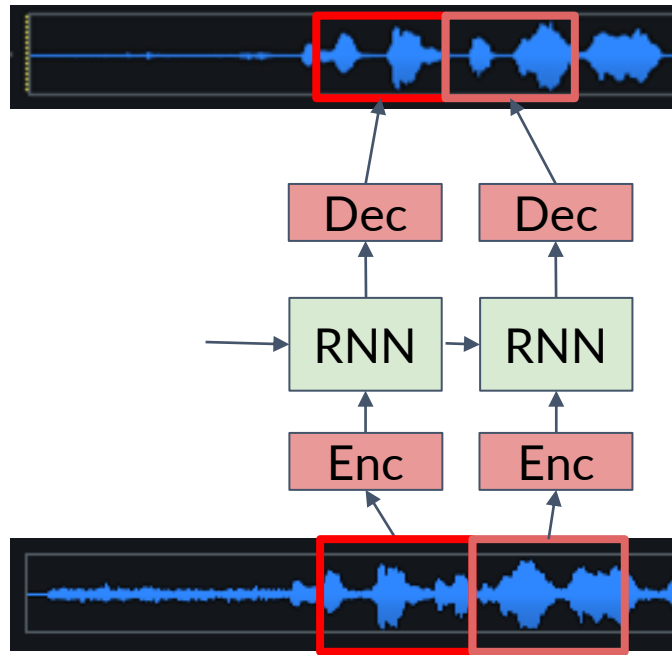
DEMUCS Denoiser (2020)



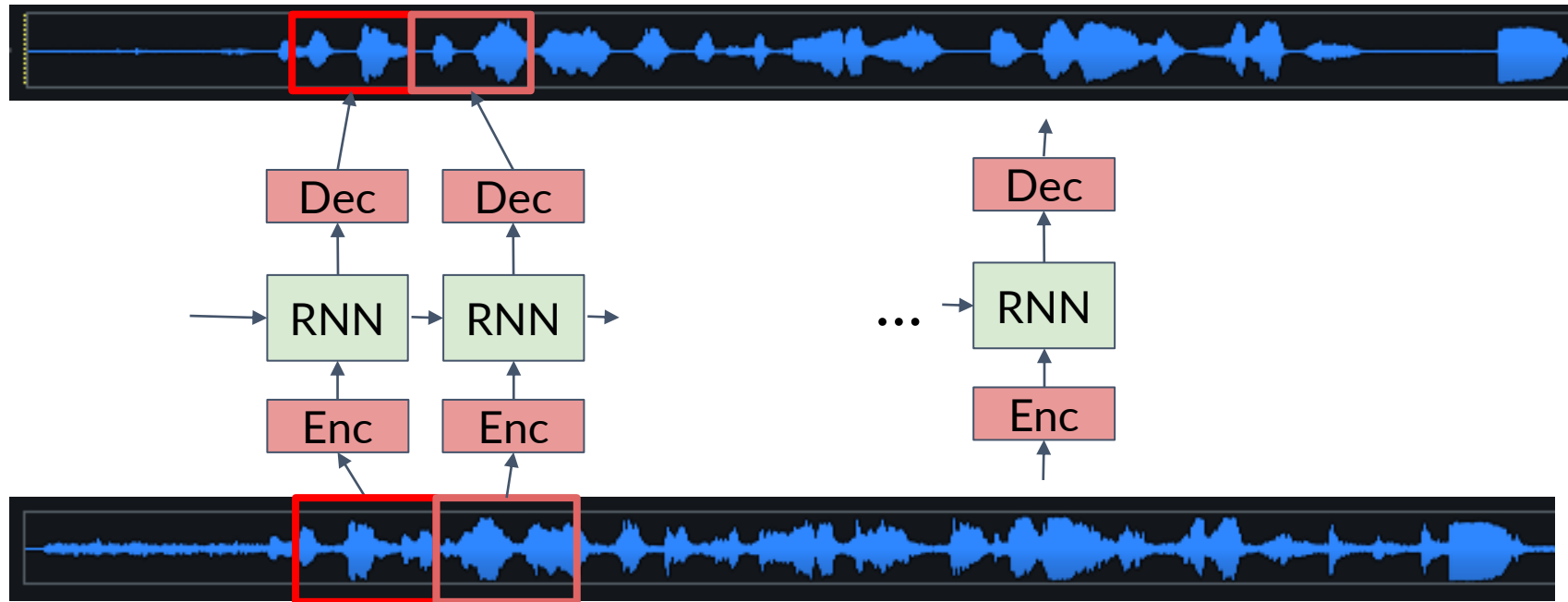
DEMUCS Denoiser (2020)



DEMUCS Denoiser (2020)



DEMUCS Denoiser (2020)



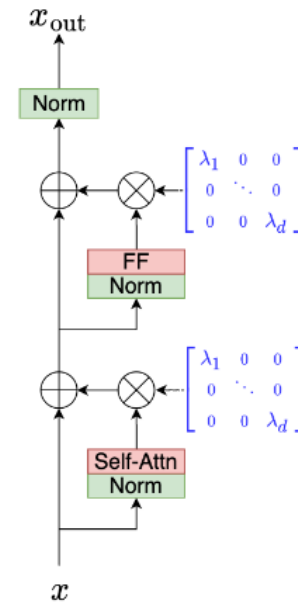
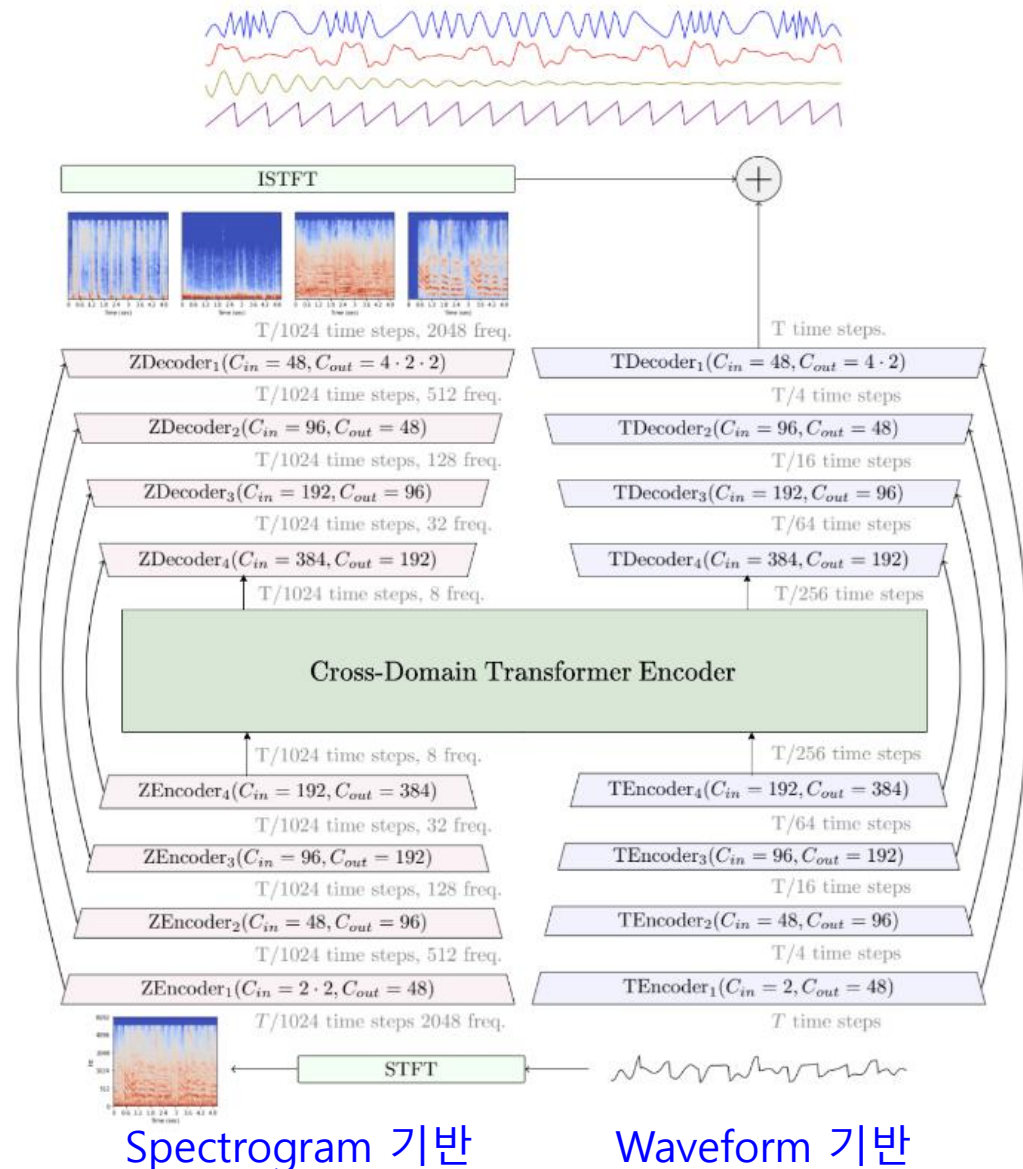
DEMUCS Denoiser (2020)

Architecture	Wav?	Extra?	Test SDR in dB				
			All	Drums	Bass	Other	Vocals
IRM oracle	✗	N/A	8.22	8.45	7.12	7.85	9.43
Wave-U-Net	✓	✗	3.23	4.22	3.21	2.25	3.25
Open-Unmix	✗	✗	5.33	5.73	5.23	4.02	6.32
Meta-Tasnet	✓	✗	5.52	5.91	5.58	4.19	6.40
Conv-Tasnet [†]	✓	✗	5.73 $\pm$.10	6.02 $\pm$.08	6.20 $\pm$.15	4.27 $\pm$.03	6.43 $\pm$.16
DPRNN	✓	✗	5.82	6.15	5.88	4.32	6.92
D3Net	✗	✗	6.01	7.01	5.25	4.53	7.24
Demucs [†]	✓	✗	6.28 $\pm$.03	6.86 $\pm$.05	7.01 $\pm$.19	4.42 $\pm$.06	6.84 $\pm$.10
Spleeter	✗	$\sim$ 25k*	5.91	6.71	5.51	4.55	6.86
TasNet	✓	$\sim$ 2.5k	6.01	7.01	5.25	4.53	7.24
MMDenseLSTM	✗	804	6.04	6.81	5.40	4.80	7.16
Conv-Tasnet ^{††}	✓	150	6.32 $\pm$.04	7.11 $\pm$.13	7.00 $\pm$.05	4.44 $\pm$.03	6.74 $\pm$.06
D3Net	✗	1.5k	6.68	7.36	6.20	5.37	7.80
Demucs [†]	✓	150	6.79 $\pm$.02	7.58 $\pm$.02	7.60 $\pm$.13	4.69 $\pm$.04	7.29 $\pm$.06

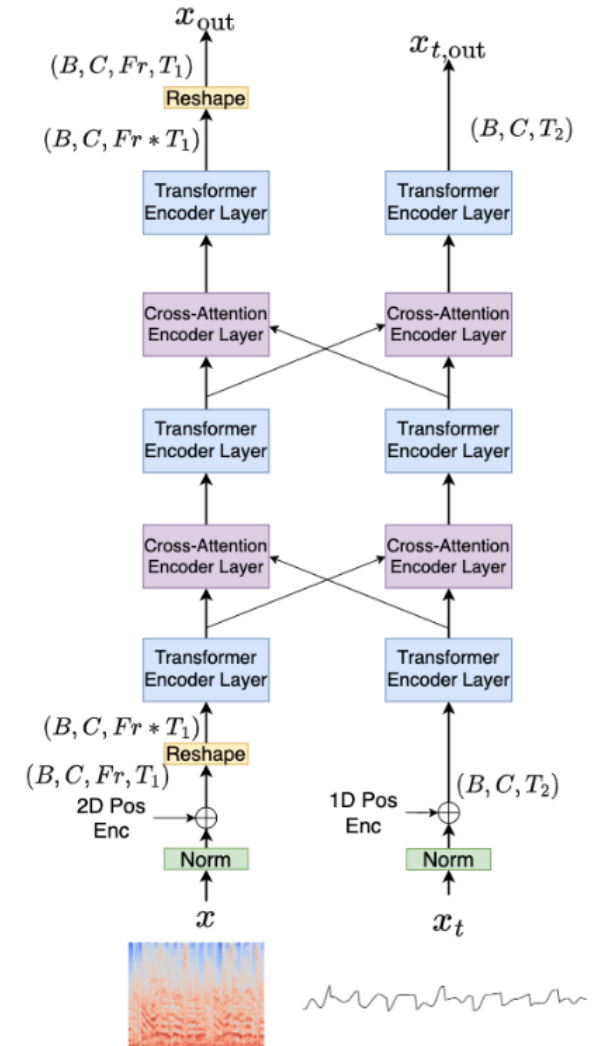
*: each track is only 30 seconds, †: from current work, ††: trained without pitch/tempo augmentation, as it deteriorates performance.

※ 위 표는 음악 분리용 Demucs (v2) 의 MUSDB18 결과 (Défossez et al., arXiv:1911.13254)
Denoiser 성능은 Valentini / DNS 데이터셋에서 PESQ · STOI 로 평가 (arXiv:2006.12847)

Hybrid Transformer DEMUCS (2022)



(a) Transformer Encoder Layer

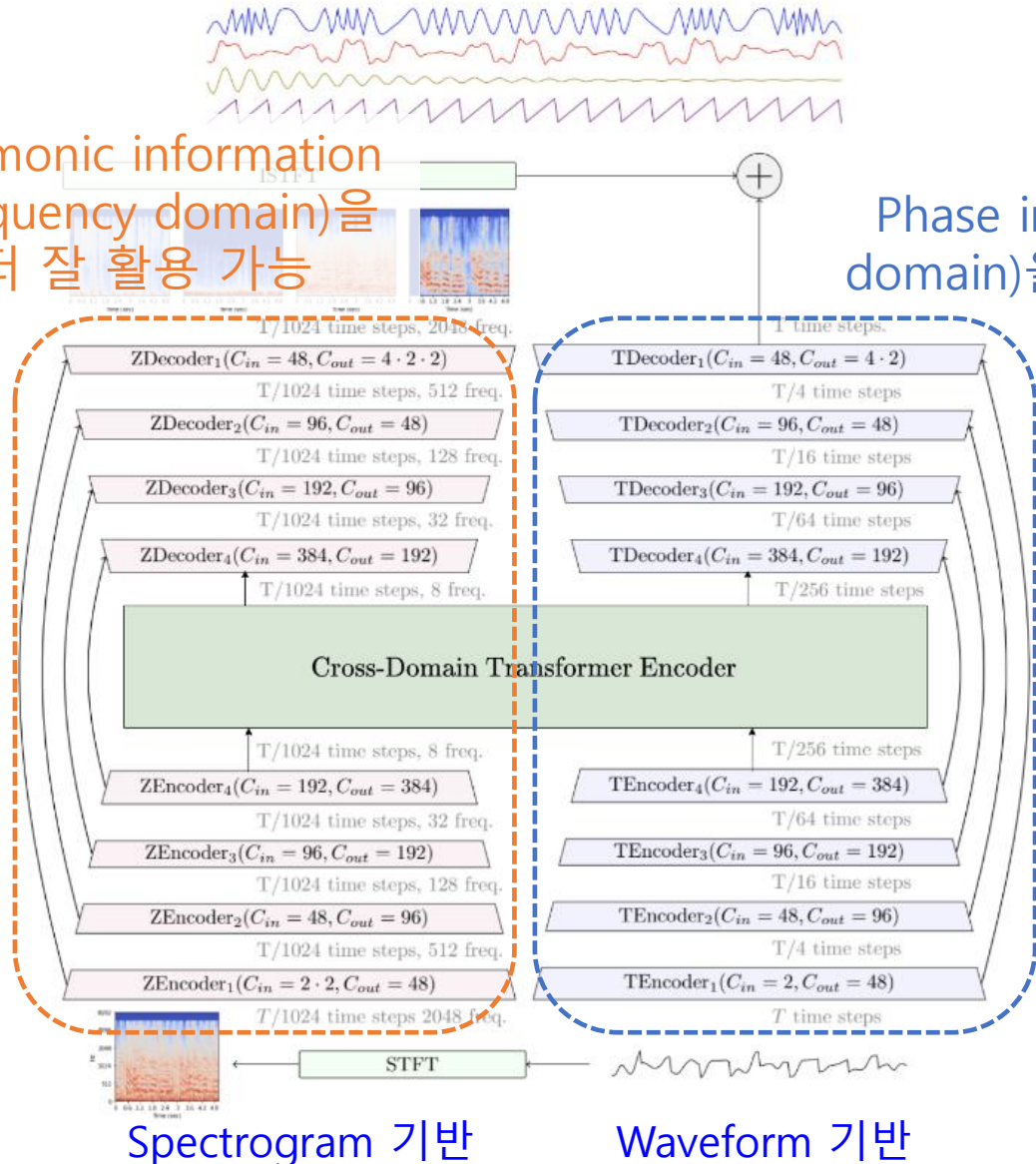


(b) Cross-domain Transformer Encoder of depth 5

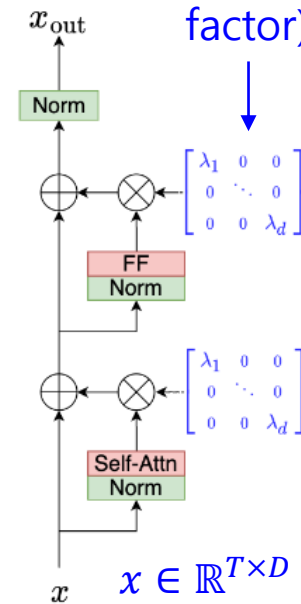
Hybrid Transformer DEMUCS (2022)

Harmonic information
(frequency domain)을
더 잘 활용 가능

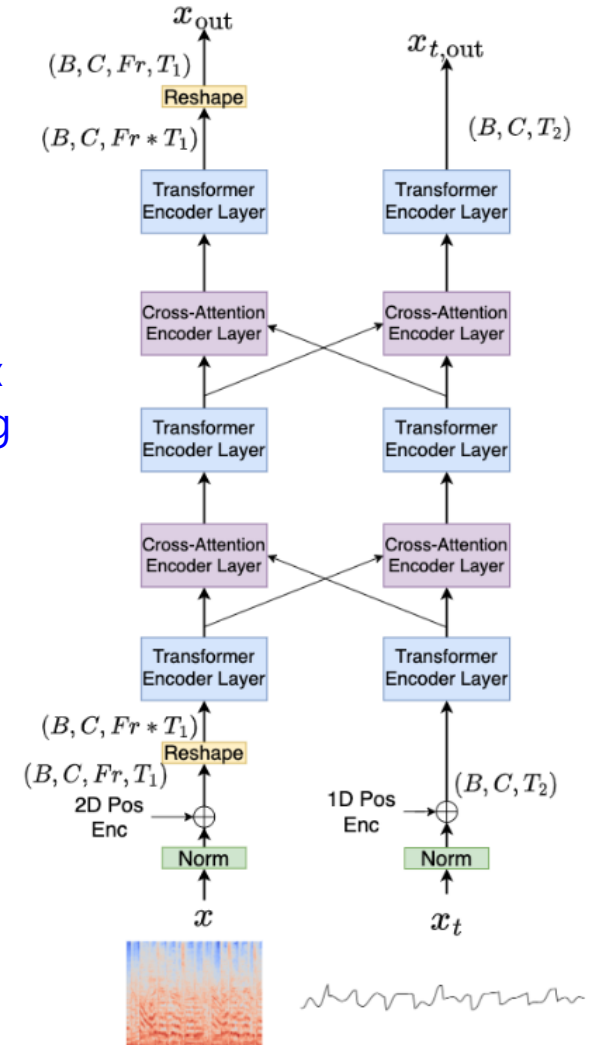
Phase information (time
domain)을 더 잘 활용 가능



Diagonal matrix
(adaptive scaling
factor)



(a) Transformer
Encoder Layer



(b) Cross-domain Transformer Encoder of
depth 5

Hybrid Transformer DEMUCS (2022)

Table 3: Comparison on the MusDB (HQ for Hybrid Demucs) test set, using the original SDR metric. This includes methods that did not participate in the competition. “Mode” indicates if the waveform (W) or spectrogram (S) domain is used. Model with a “*” were evaluated on MusDB HQ.

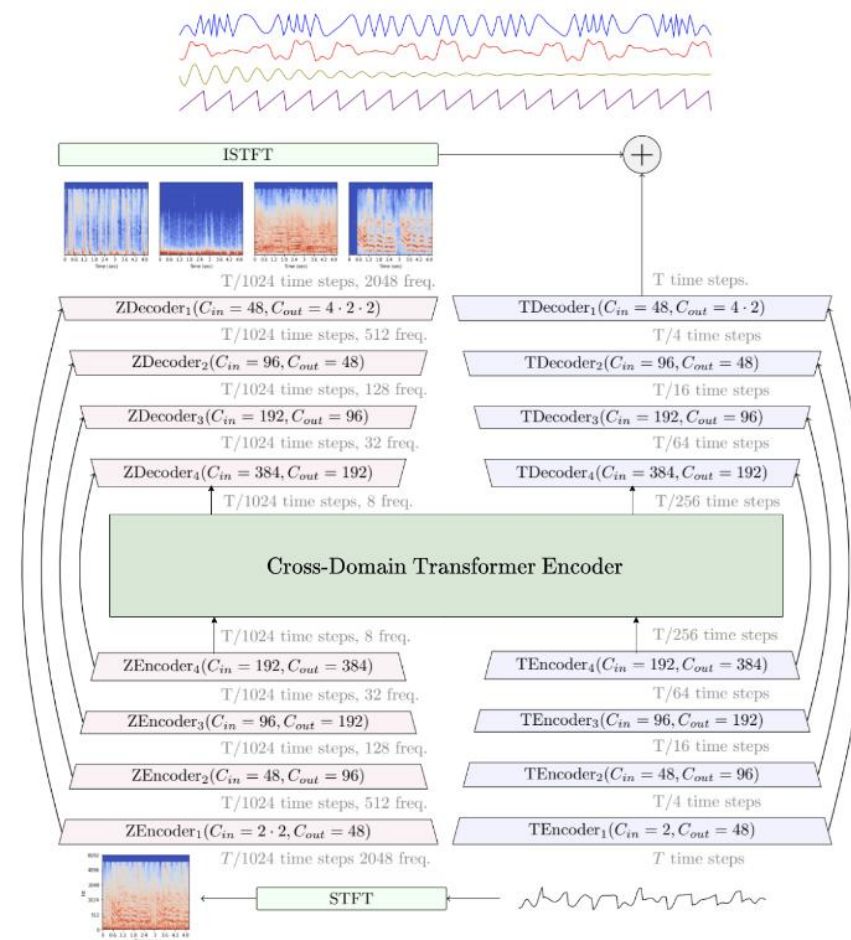
Method	Mode	All	Drums	Bass	Other	Vocals
Hybrid Demucs*	S+W	7.68	8.24	8.76	5.59	8.13
Demucs v2	W	6.28	6.86	7.01	4.42	6.84
KUIELAB-MDX-Net*	S+W	7.47	7.20	7.83	5.90	8.97
D3Net	S	6.01	7.01	5.25	4.53	7.24
ResUNetDecouple+	S	6.73	6.62	6.04	5.29	8.98

※ 위 표는 Hybrid Demucs (v3, 2021) 결과 (Défossez, arXiv:2111.03600)

HT Demucs (v4) 는 MUSDB HQ 에서 9.00 dB SDR, sparse attention + fine-tuning 시 9.20 dB (arXiv:2211.08553)

BandSplit-RNN (2023)

- **More Conv, LSTM, ... ?**
 - DEMUCS has various versions: from **40M** to **86M** parameters
- Release version is highly optimized
- Reducing complexity through structured design: BSRNN (2023)

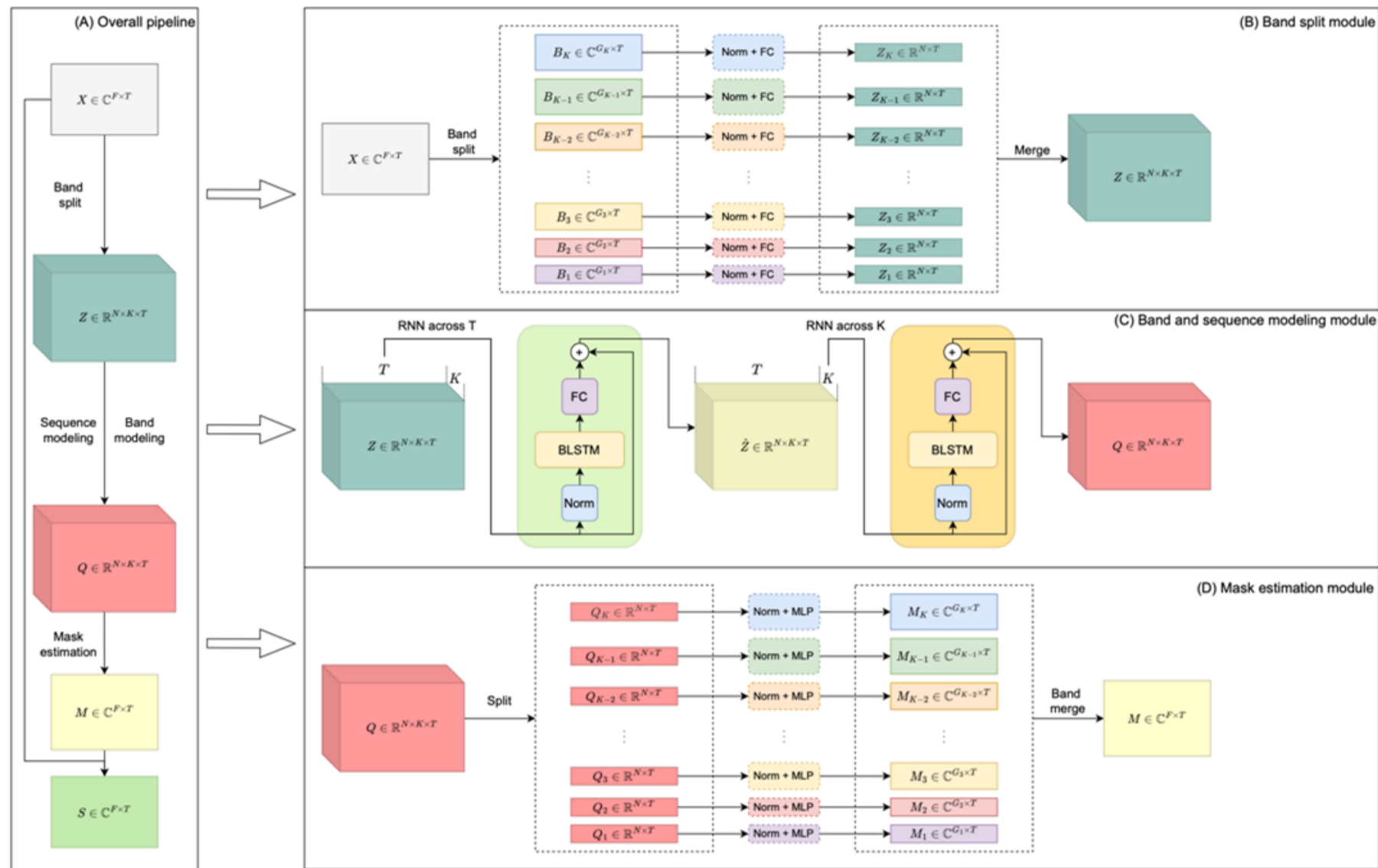


BandSplit-RNN (2023)

TF Spectrogram
(complex) split into
frequency bands

The bands are
processed by Dual-
Path RNN

Masks are
reconstructed with
band-wise MLP



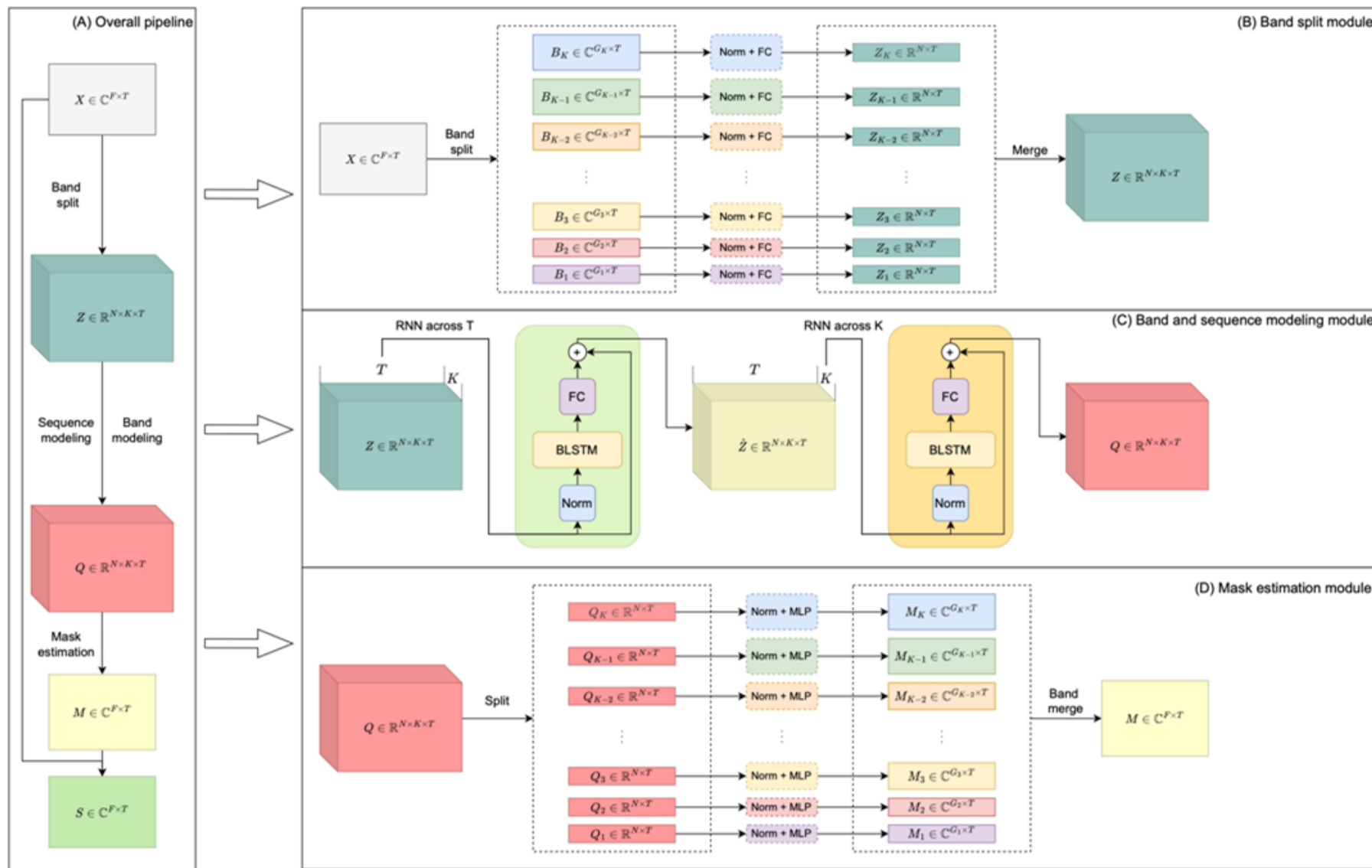
BandSplit-RNN (2023)

Supervised training + Semi-supervised finetuning

TF Spectrogram
(complex) split into
frequency bands

The bands are
processed by Dual-
Path RNN

Masks are
reconstructed with
band-wise MLP



BandSplit-RNN (2023)

TABLE III. COMPARISON WITH EXISTING MODELS ON MUSDB18-HQ (HQ) AND MUSDB18 (NHQ) DATASET.

Model	Vocals				Bass				Drum				Other				All			
	uSDR		cSDR		uSDR		cSDR		uSDR		cSDR		uSDR		cSDR		uSDR		cSDR	
	HQ	nHQ	HQ	nHQ	HQ	nHQ	HQ	nHQ	HQ	nHQ	HQ	nHQ	HQ	nHQ	HQ	nHQ	HQ	nHQ	HQ	nHQ
ResUNetDecouple+ [25]	–	–	–	8.98	–	–	–	6.04	–	–	–	6.62	–	–	–	5.29	–	–	–	6.73
CWS-PResUNet [26]	–	–	8.92	–	–	–	5.93	–	–	–	6.38	–	–	–	5.84	–	–	–	6.77	–
KUIELab-MDX-Net [32]	–	–	8.97	9.00	–	–	7.83	7.86	–	–	7.20	7.33	–	–	5.90	5.95	–	–	7.47	7.54
Hybrid Demucs [31]	–	–	8.13	8.04	–	–	8.76	8.67	–	–	8.24	8.58	–	–	5.59	5.59	–	–	7.68	7.72
BSRNN	10.04	9.92	10.01	10.21	6.80	6.77	7.22	7.51	8.92	8.68	9.01	8.58	6.01	5.97	6.70	6.62	7.94	7.84	8.24	8.23
+ finetuning	10.47	10.36	10.47	10.53	7.20	7.17	8.16	8.30	9.66	9.46	10.15	9.65	6.33	6.27	7.08	7.00	8.42	8.32	8.97	8.87

~37M params per channel against 80M of DEMUCS